# VOLLEY: Design and Computational Feasibility Assessment of a Commanded Electromagnetic CubeSat Deployment Mechanism

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***Abstract*—Rideshare CubeSats inherit a host orbit; a commanded release impulse could tailor their initial orbital states. This paper evaluates a twelve-payload 3U electromagnetic deployer through coupled force, energy, CAD, mass and matched-mission studies. The historical periodic-force model predicts 16.029 m/s, whereas finite three-dimensional geometry limits ideal work to 1.042 kJ and gives a 12.448 m/s geometry-only screen. An independent two-dimensional finite-element calculation gives 1.082 kJ and supports the end-force decline. Neither result establishes a component-rated motor. The modeled 126.6 kg dry mass gives 10.547 kg per carried 3U, exceeding the declared mass criterion and an approximate 6 kg spring-canister reference. Native CAD exposes an 11 mm transverse shortfall and multiple exact-solid interferences. No option closes the original bounded twelve-shot matched mission. These findings reject the evaluated Gen5 configuration for the stated case. A separately controlled Gen6 study selects a guided lead-screw feeder with dedicated cells, fixed 1U spacers and one indexed receiver. Its rationale combines a safety-weighted architecture trade, nominal geometry, analytical sizing and executable sequence checks. The small ranking difference is sensitive to the adopted weights. Earlier unconverged rigid-contact results are retained and supply no precision or contact-load rating. The contribution is a reproducible system-level feasibility assessment and a bounded feeder design decision; no physical prototype, measured release performance or flight qualification is claimed.**

***Index Terms*—CubeSat deployment, electromagnetic launch, Halbach array, linear synchronous motor, rideshare, constellation phasing, small launch vehicles.**

## I. Introduction

Rideshare secondary payloads may be released into an orbit selected by the primary mission rather than their own target [3]. Conventional spring dispensers supply a small, payload-dependent separation impulse; flight-proven examples include J-SSOD and commercial CubeSat deployers [2], [20]. An operator needing different delivered states can also use host manoeuvres, onboard propulsion, differential drag or an orbital-transport service. A useful deployer comparison must hold payloads, starting orbit, final-state tolerances, deadline, integration burden and failure consequences constant. Repeatable small-satellite fleets are a plausible mission context, but no customer, payload or host has endorsed the Gen5 requirements in this paper.

Electromagnetic release is established prior art, not a novelty claim in itself. Feng *et al.* model a high-acceleration induction launcher for a 20 kg CubeSat [27]; other work studies stacked-spacecraft transfer and reports a transport-mechanism prototype [30], [28], [29]. The question investigated here is narrower: can a shared host-mounted device impart a modelled, individually commanded impulse to an unmodified 3U payload while meeting a chosen acceleration limit and system-level mass, power, containment and mission criteria? The 25 $g$ ceiling is a design choice, not a universal CubeSat qualification limit; the CubeSat Design Specification defers acceptance to the mission provider [23].

This paper evaluates one fixed Gen5 computational configuration. It contributes a modeled machine and shot, selected field and circuit cross-checks, a mass and orbit trade, a reference assembly with an explicit packaging failure, and a register of conditions that prevent a product or flight claim. Host examples are reference calculations, not approved interfaces. The current results include negative design findings rather than a demonstrated customer benefit. The historical shot is retained as a model result; a new finite-stator screen challenges its use as a defensible performance prediction.

## II. Related Work

### A. Deployment heritage

The P-POD established the rail-and-spring standard, and the ISS-based J-SSOD and NanoRacks systems industrialized it [2], [3]. The failure record is instructive: the NanoRacks ball-lock anomaly, in which jack-screw preload above 0.11 N·m drove the release mechanism toward seizure, traced to a load path in which ascent preload passed through the release device itself [1]. Zhao *et al.* built a multi-CubeSat electromagnetic transport-and-ejection prototype and report selectable separation speeds of 0.4–0.85 m/s for 3U and 0.67–1.42 m/s for 1U payloads [30]. Programmable electromagnetic release and stacked feed are therefore prior art. Gen5 has a historical 16.029 m/s 3U periodic-force shot prediction within a different modeled installed-system design; its finite-geometry screen challenges that speed, and it has no measured shot and fails its own 3U mass and side-fed packaging criteria.

### B. Electromagnetic launch and supporting technology

Coilgun research examined nanosatellite launch at km/s scales [4], [5], establishing contactless acceleration and its costs: the single-stage reluctance machines of that lineage convert 1–2% of stored energy into kinetic energy, though

multi-stage induction designs do far better, and the on-orbit CubeSat launcher of [27] reports 14.9–19.9%. Ironless permanent-magnet linear machines occupy the opposite corner of the design space: the Halbach array concentrates flux on one working face [6], and the Inductrack program demonstrated large forces from such arrays with no iron and no sliding contact [8]. Finite Halbach arrays can introduce end-effect thrust ripple [7], motivating a position-dependent 3-D force check for Gen5 rather than an imported correction factor. Space-oriented supercapacitor studies and component options are reported in ESA-led programmes [9], [10], but no installed Gen5 cell string is qualified. Eddy-current damping has aerospace precedent [11]; the proposed VOLLEY brake has not been tested. On the astrodynamics side, Planet Labs demonstrated propulsion-free constellation phasing by differential drag over campaigns of weeks to months [12], [13]; that timescale is the benchmark against which impulsive seeding is measured. ISRO's POEM programme converted the spent PSLV fourth stage into a stabilized hosted-payload platform across four missions, closing POEM-3 with a controlled, debris-free reentry [14], [15], providing a hosted-stage precedent, not regulatory approval for VOLLEY.

## III. System Architecture

### A. Requirements

The Gen5 study objectives were to eject an unmodified 3U CubeSat without a customer-side armature or electrical connection, bound acceleration by a self-imposed 25 $g$ design target, command different release velocities, carry twelve payloads, and provide inhibit and abort functions. These objectives are not demonstrated capabilities. The CAD uses an ESPA-like reference ring, but no provider-approved mounting interface, host accommodation, payload dispersion budget, autonomous sequencing implementation or negligible-cost host recoil response has been established (Sec. XVI). A later Gen6 service study instead uses a generic maneuvering orbital-transfer host with a provisional parametric adapter. Its approximately 500 km sun-synchronous orbit is a vehicle-agnostic surrogate, not a selected launch provider.

### B. Derivation of Principal Design Parameters

The values that define the machine combine internal requirements, assumptions and externally sourced limits; Table I records the governing relation for each. Three deserve explicit statement. The acceleration ceiling is chosen, not derived. An earlier version of this work obtained it by integrating the NASA GEVS [21] protoflight random-vibration spectrum to 14.1 $g_{\mathrm{rms}}$, taking a $3\sigma$ envelope of roughly 42 $g$ as the launch load, and setting a 25 $g$ cap beneath it. That derivation is withdrawn. Broadband random vibration and a 162 ms commanded acceleration are different load cases: $g_{\mathrm{rms}}$ is the root-mean-square of a stationary random process over a spectrum, and multiplying it by three produces a peak-response estimate for that process, not a quasi-static limit load a structure is qualified to carry. Neither GEVS nor the CubeSat Design Specification [23] publishes a universal quasi-static level for CubeSats; the CDS defers test levels to the launch provider, and levels therefore vary by vehicle and by mission. 25 $g$ is retained here as an internal design requirement, the number the retention chain, the cradle, the abort logic and the payload-family table are all sized against, and it is not offered as evidence that any satellite is qualified for it. An earlier version of this design was acceleration-limited: at a 4.9 kg parametric sled the ejection ran at 16.3 $g$, close enough to the cap that the cap set the velocity through $v_{\max} = \sqrt{2a_{\max}L}$. That is no longer the binding constraint. With the sled computed at 9.4 kg from the CAD solid volumes the same commanded force accelerates a heavier moving mass, so the ejection sits at 10.1 $g$ and the 1.3 m zone yields 16.0 m/s. The machine is now *thrust-and-mass* limited rather than acceleration-limited, with a factor of 2.5 of margin against its own ceiling left unused. Recovering velocity therefore means removing sled mass or raising sheet current, not shortening the stroke; a 0.2 m coast-trim zone completes the 1.5 m axis. Stroke buys velocity only as $\sqrt{L}$, so the trade is unattractive: total track lengths of 1.0, 1.2, 1.5 and 1.8 m give 12.6, 14.1, 16.0 and 17.8 m/s respectively, and the envelope is already the binding packaging constraint (Sec. XVI). The capacitor bank follows from the shot energy and the droop it actually reaches: at the 2.78 kJ draw the selected 6 F bank sags to $V_1 = 0.947V_0$, and

$$C = \frac{2E}{V_0^2 - V_1^2} = 6.00\ \mathrm{F} \rightarrow 6\ \mathrm{F}, \tag{1}$$

realized as thirty-two series 190 F, 3 V commercial cells for a 96 V rail compatible with 1200 V SiC devices at a comfortable derating.

TABLE I
Governing relation for each principal parameter

| Parameter | Value | Set by |
|---|---|---|
| Accel. ceiling | 25 $g$ | *chosen* design limit (Sec. III-B) |
| Track length | 1.5 m | $v = \sqrt{2aL}$ + coast zone |
| Rail material | 7075-T6, hard anodize | CDS rail spec [23] |
| Bank voltage | 96 V | SiC 1200 V derated 60% |
| Bank capacitance | 6 F | droop eqn (2), $\leq$5% SoC sag |
| Sled magnets | N45SH | thermal $\times$ field (Sec. V-F) |
| Brake cap | 200 $g$ sled | magnet bond MoS (Sec. III-F) |
| Abort point | 45% stroke | braking-distance equality |
| Magazine | 12 (2$\times$6) | cassette CoM symmetry |

### C. Topology Trade

The payload settles the trade before the electromagnetics do. Exit velocity is bounded by

$$v_{\max} = \sqrt{2\,a_{\max}L}, \tag{2}$$

so at the 25 $g$ design ceiling over a 1.3–2 m stroke the reachable range is 25–31 m/s. The coilgun's defining advantage, a km/s-class ceiling, is outside this Gen5 payload load envelope, while its costs remain: microsecond timing against the suck-back effect, an armature bolted to the customer's satellite, and no abort once fired. Efficiency is deliberately not among them: at 14.9–19.9% [27] a multi-stage coilgun is comparable to the figure reported here, and an earlier version

TABLE II
TOPOLOGY TRADE. THE VELOCITY ROW IS EVALUATED AT THIS DESIGN'S CHOSEN 25 $g$ CEILING, NOT AT A QUALIFICATION LIMIT OF THE PAYLOAD

| Criterion | Reluctance coilgun | Ironless LSM + sled |
|---|---|---|
| Velocity at the 25 $g$ ceiling | 25–31 m/s | 25–31 m/s |
| Electrical→payload efficiency | 14.9–19.9% [27] | 18.8% (Sec. V) |
| Satellite modification | armature required | none |
| Velocity control | pulse timing | closed-loop servo |
| Abort | none once fired | to ∼45% stroke |
| Field on stored satellites | pulsed, unshielded | Halbach self-shielded |

of this argument claimed otherwise on the strength of a single-stage number. A linear synchronous motor concedes nothing in the reachable envelope and inverts every cost: high drive efficiency, continuous servo control, and a reusable sled that carries the magnets so the satellite carries nothing. Table II summarizes.

### D. Layout and Firing Cycle

The model places the release station 1.5 m from the breech on 1.8 m structural longerons; the powered stroke is 1.3 m, followed by a 0.2 m coast-and-trim interval (Figs. 1 and 2). The stator is a double-sided ironless three-phase winding. The modeled mover is a 9.445 kg sled carrying opposed Halbach arrays (48 mm wavelength, 8 mm N45SH blocks, four blocks per wavelength) across a 12 mm winding gap. The two six-satellite cassettes are conceptual feed elements. Their current side-fed reference placement fails the width and exact-solid interference checks reported in Sec. XVI. Figure 3 shows the STEP-derived reference placement with the enclosure hidden; its surface shading is a visualization rather than hardware evidence.

The intended sequence is follower advance, restraint engagement, motor acceleration, sled braking and payload departure. Feed forces, contact transitions, retention during ascent, release timing and tip-off have not been verified for an actual satellite. The current model assumes that the payload remains seated against the aft backstop during acceleration and separates when the sled decelerates; that assumption requires moving-contact analysis and instrumented tests. The ideal internal mass-motion model gives no final rigid-body rate after the sled stops, but it predicts a transient return rate of 0.130 deg/s for the assumed 500 kg host and 0.385 deg/s at 200 kg; both declared peak-rate bands fail. The earlier 8.2 s rate-null and 18.1 s cadence floor are superseded. An assumed 166 mm lever arm gives a 0.40 deg attitude offset in the 500 kg example. Controller restoration time and structural ringing have not been established. The adopted mission-planning interval is 1200 s between shots; 10–20 s is only a modeled recharge floor, so twelve scheduled shots span approximately 4.0 h and 2.6 orbits if all other operations succeed.

### E. Restraint, Inhibits, and Abort

Launch restraint separates the preload path from the release path, the inverse of the NanoRacks fault [1]. One-shot retention gates at each cassette exit carry ascent preload directly

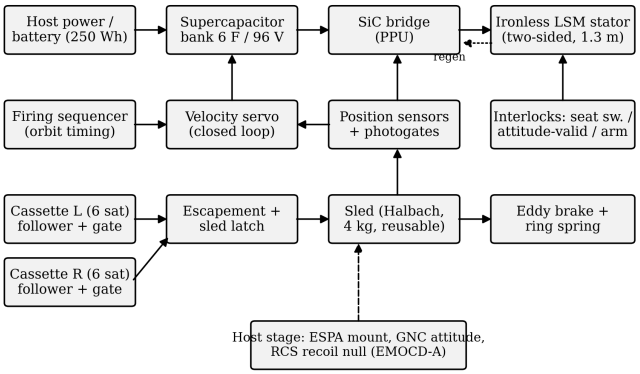



Fig. 1. System block diagram: power, control, mechanism, and host-interface chains.

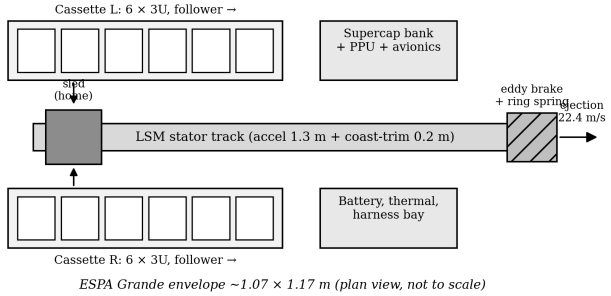



Fig. 2. Plan-view layout against an ESPA-Grande-class envelope (not to scale), retained as a dimensional reference. The closed envelope exceeds that class in length by 44%; ESPA-Grande port compliance is not a requirement of this design, and the target host class is discussed in Sec. XVI.

into structure; the escapement is caged during ascent; the sled is held by a motorized over-center cam lock. Firing requires a three-inhibit chain: redundant seat switches, an attitude-valid flag from the host, and a sequencer arm. Abort is available to approximately 45% of the stroke, beyond which braking distance for the combined mass exceeds remaining track; the muzzle buffer is sized to capture an aborted combined mass. The retention gate carries the ascent preload of a six-satellite stack through two D9 A-286 shear pins of 41.0 kN double-shear capacity; the escapement, caged behind the gate, sees none of this. The quasi-static case taken at this design's own 25 $g$ ceiling, $24\,\mathrm{kg} \times 25\,g = 5.9$ kN, is *not* the sizing case, and the gate was originally sized against it with two D6 pins. Miles' equation on the GEVS protoflight spectrum (14.1 $g_{\mathrm{rms}}$, 0.16 $g^2$/Hz) at the track's 109 Hz fixed-fixed mode gives a $3\sigma$ load of 11.7 kN at $Q = 10$ and 20.2 kN at $Q = 30$, two to three and a half times the quasi-static figure, which left the D6 margin negative at $Q = 30$. The pins were not undersized; the load case was. Resizing to D9 raises the margin of safety to $+0.45$ at $Q = 30$ and keeps it positive across $Q = 10$–$30$ for an added mass of 11 g, so the retention chain no longer depends on where the unmeasured structural damping lands, which matters because the same unmeasured $Q$ governs the

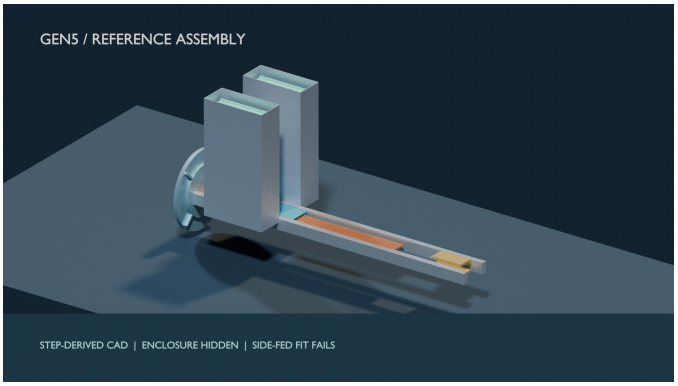



Fig. 3. Twenty-instance Gen5 reference assembly visualized in Blender after FreeCAD read-back of named STEP solids. The enclosure is hidden to expose the track and side cassettes. Their evaluated placement has a documented exact-solid interference; the image is geometry visualization, not a built or qualified mechanism.

force-ripple resonance discussed in Sec. XVI and cannot be resized away there. Random vibration, not quasi-static preload, is the driving case for the retention chain, and is now the case it is sized against. The abort threshold is where the sled braking distance under the same thrust equals the remaining track, which for the loaded 13.4 kg carriage falls at 45% of stroke and 11.1 m/s. The commit fraction is unchanged by the heavier sled, braking distance and travelled distance scale with mass identically, so the mass cancels, but the velocity at the commit point falls with the acceleration.

### *F. Arrest, Power, and Materials*

In the historical periodic-force case, the sled leaves release with about 1.21 kJ. Motor regeneration alone cannot arrest it, because braking force is bounded by the same thrust constant as acceleration; that bound sets what regeneration can and cannot do, and both halves matter. Braking at the rated sheet current over the 39 mm of stator the closed envelope allows downstream of release extracts 51.8 J of mechanical work for 0.4 J of copper, returning 47 J to the bank, and the sled still reaches the brake at 15.7 m/s carrying 1162 J. A copper-fin eddy-current brake therefore provides bulk arrest: force proportional to velocity, contactless, with flight heritage in the damper class [11]. The historical brake model assumes a tapered pole entry and a 200 $g$ sled-deceleration cap; its achievable force and heat rejection have not been independently established. This limit is set by the magnet retention: a 65 g Halbach block at 200 $g$ loads its bonded interface (10.8 cm$^2$) to only 0.12 MPa against a conservative 10 MPa epoxy allowable, a margin of 84, so 200 $g$ is a deliberate conservatism protecting the brittle sinter rather than a strength limit. The 0.344 kg copper fin absorbs the sled's residual 1162 J per shot at an adiabatic 9.0 K per-shot transient rise; cooling between shots has not been modelled; a ring-spring catches the residual below 1.5 m/s (Fig. 4). Pulse energy comes from a 6 F, 96 V supercapacitor bank switched by a SiC bridge [9]. Materials follow three rules: nothing conductive rides the changing field (titanium sled chassis, PEEK formers),

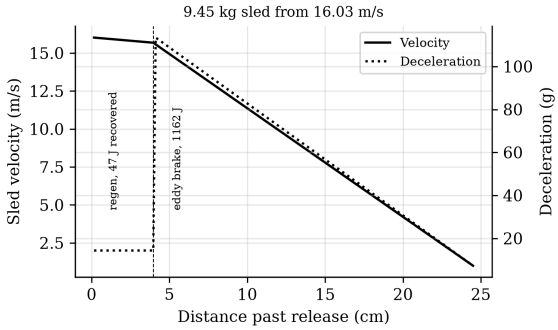



Fig. 4. Eddy-brake arrest, taper-limited to 200 g on the sled.

nothing near the track is ferromagnetic except by design (BeCu springs, A-286 fasteners), and nothing outgasses near customer optics (E595-compliant polymers, dry films, sealed roller cartridges).

## IV. Models and Verification

### *A. Field Model and Independent Verification*

The Halbach airgap field is a decaying wave, $B(y) = B_0 e^{-ky}$ with $k = 2\pi/\lambda$, summed over the two opposed faces. Because this carries the force budget, it was checked against an independent magnetostatic computation (analytic cuboid superposition, magpylib), with the array rotation-sense convention determined empirically by probing a single array on both faces; the exercise caught two successive sign errors whose outputs matched single-array fingerprints. Converged results: single-array mid-gap 0.351 T against the wave model's 0.351 T; double-sided mid-gap peak 0.694 T against a predicted 0.702 T. The stray field behind the array back face, which sets the magnetic keep-out for stored satellites, computes to 22.7 mT at 10 mm, 4.3 mT at 20 mm, and 0.4 mT at 50 mm.

### *B. Winding-Resolved Thrust Constant*

The periodic winding calculation averages $B_y$ across the array's full 90 mm depth before applying the Lorentz integral. Its reported thrust constant changes from the superseded 11.03 to 10.54 N/(kA/m), approximately a 4.4% reduction on the rounded values. This integral correction is distinct from a ratio of field samples and does not resolve finite-stator end effects. A slotless three-phase belt winding (pole pitch $\lambda/2$, belts of $\lambda/6$, winding factor 0.955) was resolved against the verified field by direct Lorentz integration with position-locked sinusoidal commutation, the field translated with the sled. The result is a thrust constant

$$K_t = 10.54\ \mathrm{N/(kA/m)}, \quad \text{ripple } \pm 1.01\% \text{ (6th harm.)}, \tag{3}$$

shown in Fig. 6. The field map in Fig. 5 gives a centre-plane peak of 0.5041 T and a depth-mean sample of 0.4759 T. Their ratio is 0.9441; the centre-plane sample is about 5.9% greater than that mean. This pointwise field ratio is not the 4.4% integrated thrust-constant correction.

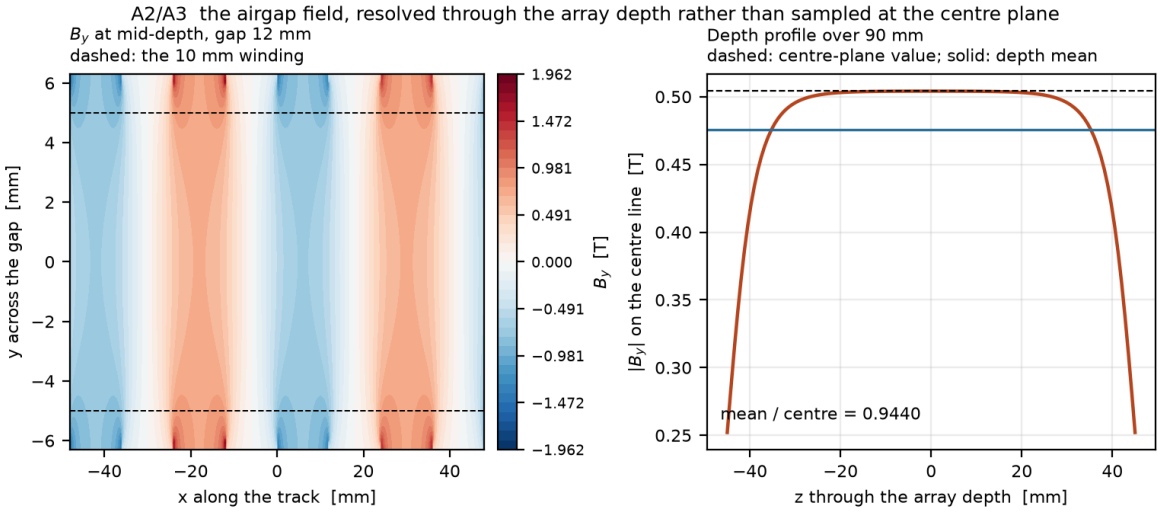



Fig. 5. Halbach airgap field. Left, $B_y$ in the $x$–$y$ plane over one wavelength; right, the profile through the array's 90 mm depth. The plot and winding integral use the same field builder; the plot is a model visualization, not an independent validation.

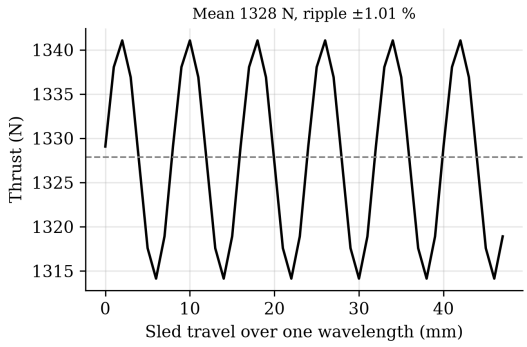



Fig. 6. Winding-resolved thrust over one wavelength of sled travel.

The periodic model includes the traveling-wave phase factor and assumes sheet-current limits of 140 kA/m with a 126 kA/m command (21 A/mm$^2$ in the represented copper). These are analytical inputs, not component ratings. The single-pulse adiabatic estimate is $854/(6.29 \times 385) = 0.3527$ K, or approximately 0.35 K, for the assumed copper mass and heat capacity. It excludes hotspots, conduction, mounting and repeated-shot duty-cycle effects.

### C. Shot, Servo, and Astrodynamic Models

The shot integrates sled-plus-satellite dynamics against the supercapacitor circuit with explicit copper loss and a 95% converter. Velocity control uses a position-scheduled profile with proportional correction and a coast-trim residual correction from photogate measurement; an 800-run Monte Carlo over thrust-constant, mass, force-ripple, and sensor tolerances gives the dispersion. The proportional gain is not tuned to that dispersion but designed against stability margins, as Sec. IV-D sets out. Orbital lifetime uses per-revolution orbit-averaged decay (Gauss tangential equations, quadrature over eccentric anomaly, static exponential atmosphere at mean activity [17]); the method was cross-validated against an independent Cowell RK4 propagation with drag, the two agreeing on 30-day semi-major-axis decay to 99.4%. Constellation seeding uses $\Delta a = 2a\,\Delta v/v$ and the resulting period difference; deployment safety propagates the stage and all deployed ellipses with Kepler-plus-secular-$J_2$ motion for 30 days, with candidate minima refined at 0.25 s sampling.

### D. Velocity-Loop Stability

The control law divides the commanded sheet current by the modelled thrust constant and multiplies by the modelled mass, so the plant's own $K_t/m$ cancels and the open-loop transfer reduces to $L(s) = K_p\, e^{-s\tau}/s$. The proportional gain is therefore not a current gain but an acceleration per unit velocity error, in $\mathrm{s}^{-1}$, and *its numeric value is the gain crossover frequency in rad/s*. The total lag $\tau$ is the sensing and computation transport delay plus the half-sample of a zero-order hold; both are pure delays and add. No sensor has been selected, so the delay is a stated assumption, swept rather than asserted, and 0.6 ms is used as the reference value with a 0.1 ms hold at 5 kHz.

That reduction sets the design constraint. An earlier gain of 3500 $\mathrm{s}^{-1}$ places the crossover at 557 Hz, five times the track's 109 Hz first bending mode and above the 48 Hz mode as well, giving $-50.4°$ of phase margin and $-3.9$ dB of gain margin at the reference delay, with marginal stability reached at only 0.45 ms of total lag (Figs. 7 and 8). Two properties of the simulation had concealed this: the loop fed back an undelayed state, which is the one condition under which that gain is stable, and the command is clipped at the winding rating, so the linearly unstable loop behaves as a bang-bang relay whose mean follows the feedforward term while the terminal photogate trim removes the residual. The reported dispersion was consequently governed by the saturation limits

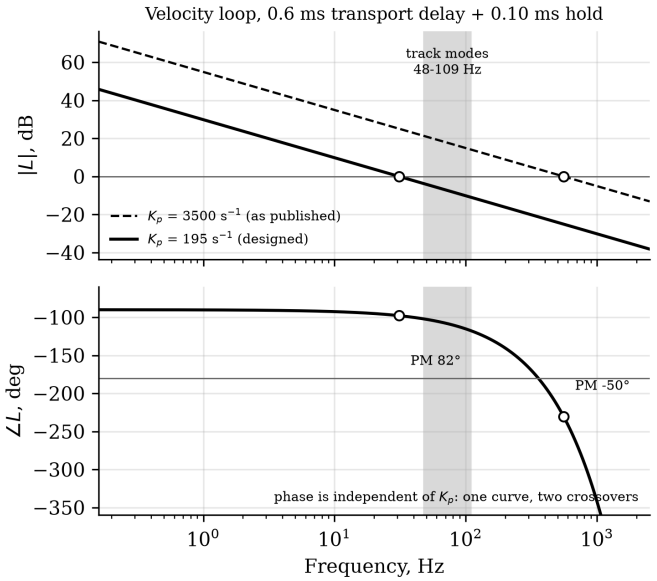



Fig. 7. Open-loop velocity-loop response at the earlier and the designed gain, with the 48–109 Hz band of track modes shaded and both gain crossovers marked. Phase depends only on frequency and delay, so the two gains share one phase curve.

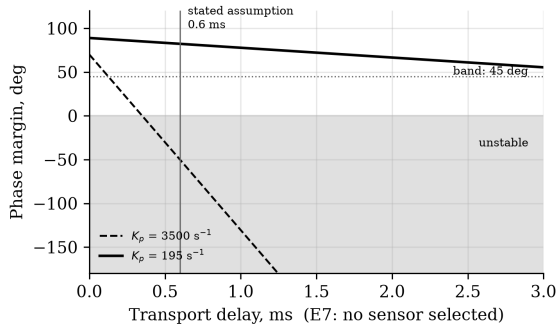



Fig. 8. Phase margin against transport delay. The earlier gain crosses into instability at 0.35 ms of transport delay; the designed gain holds more than $55^\circ$ across the whole swept range.

and the terminal correction rather than by the feedback it was attributed to, which is why the closed-loop figure is reported here only alongside the margins that make it a property of a design.

The gain is instead sized as the largest value simultaneously holding $\geq 50^\circ$ of phase margin at the reference delay and keeping closed-loop bandwidth at or below one third of the first bending mode, which gives 195 s$^{-1}$: crossover 31.1 Hz, phase margin $+82.2^\circ$, gain margin $+21.2$ dB, and no excursion above the current rating anywhere in the stroke. The gain falls by a factor of eighteen and the $3\sigma$ dispersion does not move, 0.0271 to 0.0267 m/s. The loop never required the bandwidth: dispersion is set by the terminal trim and by the thrust-constant and mass tolerances, not by loop gain. The binding constraint is now the structural one, so a measured first mode below 109 Hz would lower the designed gain with it. Classical margins on a 162 ms trajectory are necessary rather than sufficient, the plant is treated as rigid, and the structural modes enter only as a frequency the bandwidth is held away from; a compliant-track model remains outstanding.

## V. Performance Results

### A. Historical Periodic-Model Shot

The following values belong to the historical periodic-force shot and are not a verified Gen5 rating; the finite-geometry screen in Sec. XV challenges their force input. At that modeled point (3U reference, 9.445 kg sled input from historical Gen3 CAD solid volumes, 1.3 m acceleration): exit velocity 16.0 m/s at 10.1 g, pulse 162 ms, peak current 320 A, and bank state-of-charge sag 5.2% (Fig. 9). Energy drawn is 2.78 kJ per shot, with copper heat of 854 J and a further 76 J dissipated in the bank's own series resistance. Regenerative braking returns 47 J after release, so the net draw is 2.74 kJ. The payload receives 514 J, a 18.8% electrical-to-payload efficiency; the sled's share is split between 47 J recovered and 1162 J dissipated in the arrest brake. Efficiency falls with the heavier sled for two compounding reasons: a larger fraction of the same mechanical work goes into a mass that is then braked away, and the longer 162 ms pulse accrues more copper loss at unchanged current density. Recovery is what partially offsets the first of those, and it is bounded by geometry rather than by loss: over the available 39 mm the braking pulse lasts 2.5 ms and burns 0.4 J of copper, against 854 J over the 162 ms shot, so recovered energy rises monotonically with braking force up to the winding's rating and with distance beyond it. A 500 mm section would return 540 J and a 1 m section would arrest the sled outright while returning 727 J, but the closed envelope already exceeds its target class by 44% (Sec. XVI), so distance is not available here. Peak current during recovery is 212 A, below the shot's own 320 A, so the drive is not re-rated. The closed-loop Monte Carlo returns a dispersion of 0.0274 m/s ($3\sigma$) at a 15.8 m/s fleet setpoint (Fig. 10), dominated by the sensing chain rather than plant tolerances. The setpoint must sit below the open-loop ceiling for the servo to retain authority; at 98.85% of it, the nominal headroom is 0.188 m/s. Table III gives the payload family: the system is force-limited above 1U, so heavier payloads trade modeled velocity under the internal 25 $g$ ceiling; actual structural compatibility is unverified, and the deployer mass each customer pays for varies by a factor of 109 across the family (Sec. V-B).

A separate local algebraic audit recalculates the payload kinetic energy, constant-acceleration equivalent stroke, dry-mass sum and brake-work identities from the captured JSON. These agree within the declared rounding bands. The compact output leaves 124.488 J (4.47%) after those terms, now reconciled to 90.964 J assumed converter loss, 32.460 J auxiliary draw and 1.064 J forward-Euler integration excess. The rounded residual is below 0.001 J. This closes the historical model arithmetic, while an actual power source and the revised finite-force circuit remain unverified.

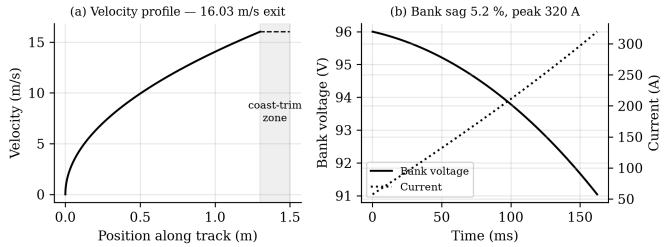



Fig. 9. Historical periodic-model shot profile: (a) velocity with coast-trim zone; (b) bank voltage and current.

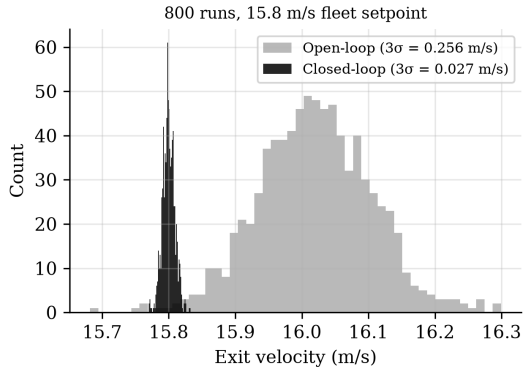



Fig. 10. Closed-loop exit-velocity dispersion, 800 Monte Carlo runs.

### *B. Payload Class and the Mass-per-Satellite Argument*

Table III is read most usefully from its last column rather than its first. Exit velocity is nearly flat across the family, because the 9.445 kg sled is most of the moving mass: removing the payload entirely buys 19% of velocity, so velocity is a property of the sled and the stroke and not of the customer. Shortening the magnet array does not recover it either, since array length sets thrust and sled mass together, a 150 mm array gives a 5.15 kg sled but a 4.65 N per kA/m thrust constant, and the 3U case falls to 12.9 m/s.

The column that does move is deployer mass per satellite: by a factor of 27 between the 3U case and PocketQube 1P, and by 109 across the full column. At 3U the machine spends 10.55 kg of deployer per customer, which loses to a cold-gas propulsion module delivering the same 16.0 m/s at 0.5–1.2 kg by a factor of between nine and twenty-one, 12.4 against the 0.85 kg mid-class figure. At PocketQube 1P it spends 0.388 kg and wins by 1.3 to 3.1. The comparison inverts on payload class, and on no machine parameter at all: not the thrust constant, not the stroke, not the sheet current. It inverts less decisively than it did: the figures published until 2026-08-20 were 7.04 and 0.24 kg, and the enclosure buildup that replaced an 8.00 kg packaging placeholder with 50.04 kg of derived line items moved both.

Three qualifications belong with those counts rather than beneath them. First, they are volumetric. The packing efficiency is calibrated against the one configuration that has been laid out in CAD, the 3U case, where a raw volume ratio

TABLE III

HISTORICAL PERIODIC-MODEL PAYLOAD-FAMILY SCREEN (1.3 M ACCELERATION, 9.445 KG CAD-DERIVED SLED). DEPLOYER MASS PER SATELLITE USES THE 126.6 KG DRY MASS AND A PACKING MODEL CALIBRATED SO THE 3U CASE RETURNS THE TWELVE THE MAGAZINE IS LAID OUT FOR. EVERY COLUMN IS REGENERATED FROM `PAYLOAD_FAMILY.JSON` AT THE CURRENT OPERATING POINT; THE VALUES PUBLISHED UNTIL 2026-08-20 PREDATED BOTH ADR-030 AND THE ENCLOSURE BUILDUP

| Class | Exit vel. | Accel. | Per load | kg/sat. |
|---|---|---|---|---|
| PocketQube 1P (0.25 kg) | 18.9 m/s | 14.0 g | 326 | 0.388 |
| ThinSat (0.28 kg) | 18.8 m/s | 13.9 g | 123 | 1.029 |
| PocketQube 3P (0.75 kg) | 18.4 m/s | 13.3 g | 108 | 1.172 |
| TubeSat (0.75 kg) | 18.4 m/s | 13.3 g | 41 | 3.088 |
| 1U (1.33 kg) | 17.9 m/s | 12.6 g | 40† | 3.165† |
| 3U (4 kg) | 16.0 m/s | 10.1 g | 12 | 10.550 |
| 6U (8 kg) | 14.1 m/s | 7.8 g | 6 | 21.100 |
| 12U (12 kg) | 12.7 m/s | 6.3 g | 3 | 42.200 |

predicts 21 and the magazine holds 12, and the resulting 56% is then applied to classes for which no cassette exists. Second, the feed engages the CubeSat Design Specification corner-rail interface at the cradle, the escapement and the retention gate; PocketQubes, ThinSats and TubeSats do not have that interface, so a smaller class needs its own cassette, cradle and gate, which is mechanical design rather than a parameter change. Third, hundreds of satellites per load is a different machine from twelve: the campaign thermal case, the bank recharge duty and the escapement cycle life were all sized for twelve shots, and the bank is already the binding constraint at that number (Sec. XVI). What Table III establishes is that payload class dominates the economic argument, not that a small-payload variant has been designed.

The clearest 3U mass comparison is with a conventional spring canister. A canisterised spring dispenser occupies roughly 6 kg per 3U slot, and an acceptance band written before the comparison script existed asked for parity within 15%. The modeled ratio is 1.758: this deployer is 76% heavier per satellite than the hardware it would displace, and the band fails. Replacing an 8.00 kg packaging placeholder with a 50.04 kg derived enclosure buildup increased the modeled device mass and caused the 15% parity criterion to fail. The project's own kill criterion on mass per satellite, fixed at an estimated 2 kg, is crossed by 5.3×. No enclosure change reaches parity: the lightest alternative structure costed, a honeycomb sandwich, would give 8.87 kg per satellite and a ratio of 1.48. The route that moves this criterion is a smaller payload class, which is what the left of Table III prices and what Sec. XVI records has not been designed.

It is not for want of looking. Every kilogram of the dry mass was attributed to the requirement that causes it, six requirements, from *the satellite is unmodified* to *twelve satellites share one drive*, and each of the 64 subsets was then deleted in turn to find the lightest machine any combination of relaxations could reach (Fig. 11). Three of the six carry no mass on their own. The best corner deletes both of the two that do, giving up the reusable mover and the requirement that

the energy arrive during the shot, and it removes 37.89 kg, or 29.90%. 88.67 kg survives every deletion in all 64 corners, 70.06% of dry mass, 7.39 kg per satellite against a 2 kg criterion. Deleting requirements does not close it; only carrying more, smaller satellites does. The ledger is an attribution and an upper bound on what relaxation buys, not a sizing model, and it does not claim the surviving mass is irreducible by better design.

The architecture does not meet the spring-dispenser mass parity criterion. A separate earlier claim that waiting between releases gives satellites a persistent phase offset at zero relative velocity is also withdrawn: co-orbital zero-impulse releases from an unchanged host share the same state at a common epoch, as Sec. V-C explains. The contribution evaluated here is a modelled, commanded per-satellite orbital impulse without modifying the satellite, at a design acceleration of 10.1 $g$. Payload structural compatibility remains unverified; its qualified load environment and a host integration review are required. The modelled deployer costs 1.758 times the mass per satellite of the spring comparator.

### C. Astrodynamic Utility

A prograde ejection raises apogee while leaving perigee at the deployment altitude, so the correct claim is a longer-lived ellipse. At 450 km with ballistic coefficient 61 kg/m$^2$, an assumed 16.0 m/s impulse gives a modeled lifetime multiplier of 1.60 at mean solar activity (base lifetimes 0.52–2.61 yr across the activity range; Figs. 12 and 13).

An earlier version of this work claimed the multiplier was invariant to two decimal places across ballistic coefficients of 40–90 kg/m$^2$ and across a fivefold density range, and offered that invariance as the result the analysis could defend. That claim does not survive an independent propagator. Run headless against MSISE90 with 20×20 gravity, lunisolar third bodies and solar radiation pressure, GMAT R2022a reproduces the multiplier at mean and high activity but returns 2.07 at low activity, a spread of 18% against a 5% acceptance band fixed before the run. The reason is a defect in how the sweep was posed rather than in the arithmetic: the model represents solar activity as a uniform multiplicative scale on density, and ballistic coefficient enters the drag term in the same multiplicative slot, so both sweeps divide the two lifetimes being compared by the same factor. Their ratio cannot move, whatever the atmosphere does. A real atmosphere changes the *shape* of the density–altitude profile with activity, the boosted orbit's apogee samples that profile some 37 km higher than the baseline's, and the ratio then does move. The multiplier is therefore quoted here at a stated activity level, and no invariance is claimed.

Differential ejection velocities put adjacent satellites on different periods: splits of 2, 5, and 10 m/s reach 30° spacing in 6.9, 2.8, and 1.4 days. Under identical density and geometry assumptions, differential drag between 3:1 area configurations achieves the same spacing in 25.0 days (Fig. 14), consistent with flown drag-phasing campaigns operating over weeks to months [12].

A common-epoch comparison must account for the actual state imparted at release. In a two-body reference case, a satellite released with zero relative velocity from an unchanged host follows the host orbit. A second identical release later follows that same trajectory; at any common time the two satellites are co-located. The host's angular travel between release events is not persistent separation between the satellites. The 1200 s inter-shot cadence is therefore a thermal and recharge assumption, not a phasing result. Conventional spring impulses, host manoeuvres and subsequent differential drift remain valid comparators and must be included in any mission-level phasing trade.

A differential deployment impulse can change orbital period and create a growing phase separation, as the preceding model shows. It does not establish a permanent fixed angular offset without later control or drift management. The differential-drag comparison in Fig. 14 is conditional on the stated area and atmosphere assumptions. A matched comparison of commanded electromagnetic shots, spring deployment, host manoeuvres and drag is still needed before asserting a phasing advantage.

A historical-input cross-check independently propagates the Cartesian post-release state for two orbital periods and recovers a 28.800775 km semi-major-axis rise at 16.029 m/s. The 0.02 m solver/identity band passes. This verifies instantaneous two-body geometry, not the 1.60 lifetime multiplier, host campaign or launcher performance. The method and exact inputs are archived in the local validation record.

The Gen5 model instead evaluates a commanded change in the deployed satellite's orbit. Its 16.0 m/s prograde shot predicts a 28.8 km semi-major-axis increase and a 1.60 lifetime multiplier at the stated mean-activity condition. A 2.5 m/s spring is also an orbital impulse and produces an 8.2% modelled lifetime extension under the same setup. Release timing without an impulse changes neither orbit; a manoeuvring host can change the initial orbit of later releases. These are different mission cases, and the quoted Gen5 benefit is model output rather than demonstrated performance.

### D. Deployment Safety

Every prograde shot leaves an ellipse whose perigee sits at the firing altitude, so the fleet re-crosses the stage's orbit by construction, and the along-track phase between a deployed satellite and the stage realigns on a period of 10.3 days for the assumed historical velocity (Fig. 15). The closest approach reached within a 30-day screening window is deliberately *not* quoted here as a safety figure: because both bodies re-cross the same altitude, that minimum is fixed by the exact along-track phase at crossing, which is acutely sensitive to ejection velocity, in the model a $\pm 2.5\%$ change about that assumed point sweeps the 30-day minimum by more than an order of magnitude. It is a near-resonant sample of the beat geometry, not a design property. Two statements survive that sensitivity and carry the safety argument. First, per-shot collision-avoidance (COLA) screening is mandatory for every ejection, because the approach geometry cannot be certified

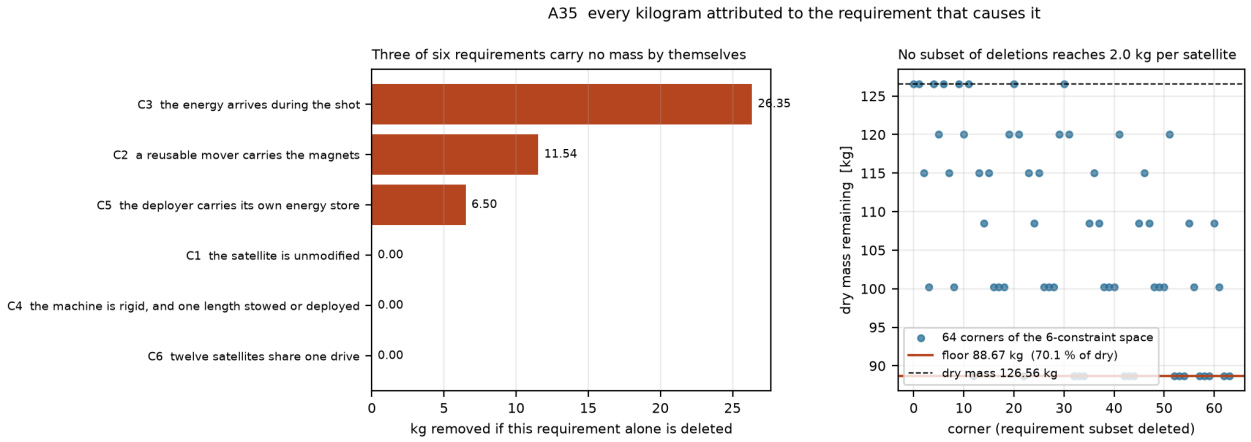



Fig. 11. Constraint ledger: dry mass attributed to the requirement causing it, and the floor reached by deleting requirements in all 64 combinations. The floor is 88.67 kg, 70.06% of the 126.56 kg rollup.

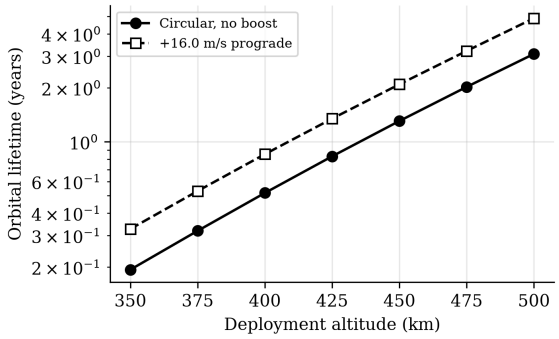



Fig. 12. Orbital lifetime vs deployment altitude (BC = 61 kg/m$^2$, mean activity).

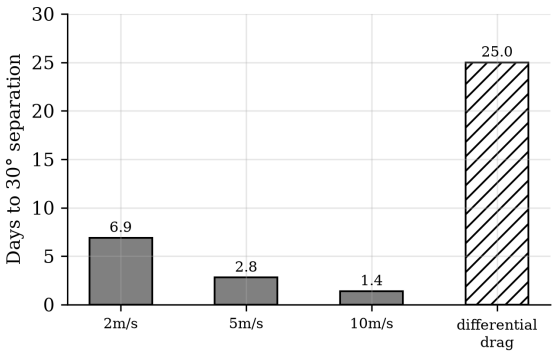



Fig. 14. Modelled velocity-split phasing versus differential drag under identical assumptions. The figure does not include spring impulses, host manoeuvres or a complete mission-level comparator.

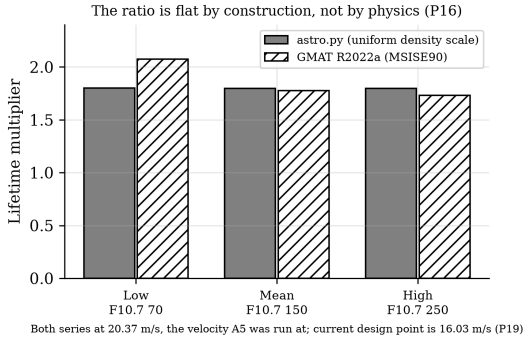



Fig. 13. Solar-activity sweep against GMAT. The static-atmosphere model returns a flat ratio by construction, not by physics; an independent propagator does not.

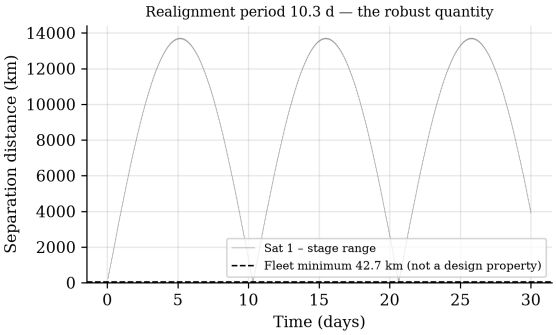



Fig. 15. Satellite–stage range over 30 days, staggered firing at the rated velocity.

from the nominal velocity alone. Second, disposing of the host stage before the first 10.3-day realignment, routine for a restartable stage, removes the recurring close-approach mechanism at its source, before the fleet can return to the stage's along-track position. Per-shot conjunction products remain a mission-operations deliverable.

### *E. Error, Tip-Off, and Mass*

Within the historical idealized control model, a 0.0274 m/s speed dispersion maps to approximately ±0.10 km in apogee at 450 km. This omits host-state and pointing uncertainty, release contact, navigation covariance and the finite-force/source discrepancy; it is not a delivered-orbit accuracy claim. Tip-off

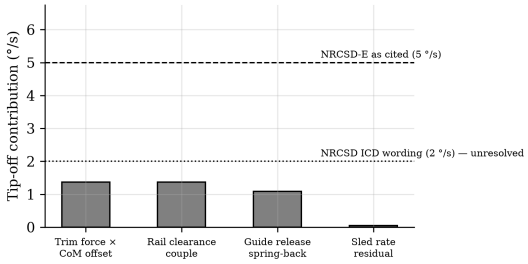



Fig. 16. Tip-off error budget against deployer requirement classes.

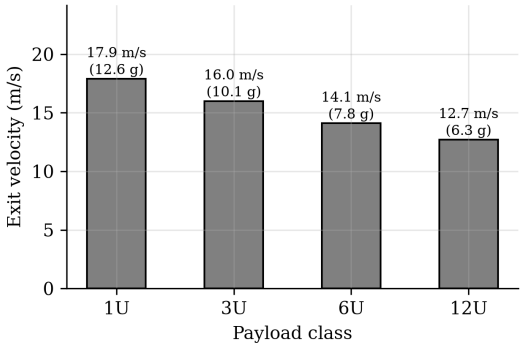



Fig. 17. Payload family: force-limited above 1U; velocity, not safety, is traded. †Counts are *volumetric*, calibrated so the 3U case returns the twelve the machine is laid out for. A subsequent fixed-cell design study replaces them where a cell geometry actually exists: 1U gives 36 per load at 2.125 kg, not 40 at 1.92, because three 100 mm units plus two dividers leave 37.5 mm of a 340.5 mm cell unusable, and ThinSat and 12U are refused outright because two of their dimensions exceed the 100 mm cell section. The binding constraint is the 166 mm cassette width.

is modelled rather than bounded, and the risk is not where the analytic budget put it. Release occurs 200 mm, 12.2 ms, into the coast-and-trim zone with commanded force already at zero, so at the residual latch and friction load the mechanism tolerates roughly 250 $\mu$s of asymmetry and still lands two orders of magnitude inside the 2 deg/s comparator. What is not benign is the start of the stroke: the payload centre of mass sits 70 mm off the thrust line, the 413 N push therefore applies a 28.9 N·m moment, and the payload accelerates across its cradle clearance at 688 rad/s$^2$, arriving at 36–231 deg/s depending on fit, 18 to 115 times the comparator. Whether that impact has damped out by release depends on a restitution model this study does not have, so tip-off is stated as a requirement on a cradle preload exceeding the 85 N couple reaction and on a release residual below about 1 N, against a release mechanism that remains undefined (Fig. 16). Recoil per shot is the satellite's momentum only, 64.1 N·s, the sled's share returning through the brake. The modeled mass rollup reports dry mass of 126.6 kg (loaded 174.6 kg, centre of gravity 0.46 m from the breech), with the ironless stator at 7.3 kg of copper and formers; the system uses roughly a third of an ESPA Grande *mass* allocation. It does not currently fit the corresponding envelope: the closed CAD assembly spans 1839 mm along the track against the roughly 1270 mm of that class, because the arrest brake sits beyond the release point and the enclosure must span it. The study therefore uses a generic maneuvering orbital-transfer host as a surrogate; no selected host class has provided an accommodation envelope or approved interface. This change in study boundary does not correct the measured ESPA-class envelope mismatch. Shortening the track to 1270 mm would leave a 731 mm acceleration zone and, since velocity goes as $\sqrt{L}$, would cut ejection from 16.0 to 12.0 m/s, a 25% loss in the quantity every performance claim here rests on. The re-scope avoids that and gives up the ESPA-Grande port population in exchange. It also leaves the packaging question unanswerable for now rather than answered: no accommodation envelope for the target host class is public, so the 1839 mm length cannot presently be shown to fit anything, and obtaining that envelope is part of the same single data exchange Sec. VI identifies. Host packaging therefore remains unverified.

TABLE IV
PER-SHOT RECOIL (64.1 N·S) VERSUS HOST CLASS

| Host class | $\Delta V$ per shot | 12-shot total |
|---|---|---|
| 200 kg free-flyer | 328 mm/s | 3.93 m/s |
| 300 kg kick stage | 219 mm/s | 2.62 m/s |
| 600 kg kick stage | 109 mm/s | 1.31 m/s |
| 900 kg (PS4 class) | 72.8 mm/s | 0.87 m/s |
| 420 t (station class) | 0.2 mm/s | 2.3 mm/s |

## VI. GENERALIZED HOST INTEGRATION

### A. Interface Requirements for Any Host

The reference host analysis assumes a mechanical mount, 150–300 W recharge allocation or an equivalent battery, an attitude reference and control authority for shot transients, and a compliant disposal plan. These are requirements to negotiate, not verified provisions of any particular stage. The quoted full-manifest momentum and propellant estimates are idealized budget calculations; actual correction, attitude recovery and disposal cost depend on host dynamics, thruster performance and the delivery sequence. Table IV varies the assumed host mass to show the translation sensitivity. A free-flyer variant would need its own GNC and is outside this Gen5 study.

### B. Worked Host Examples

ISRO's POEM missions demonstrate that a spent stage can support hosted experiments and controlled disposal [14], [15]. Published Skyroot descriptions indicate interest in restartable orbital operations [18], [19]. These are contextual examples only. No host has supplied an interface control document, confirmed available mass/power/attitude margins, accepted a magnetic payload environment or approved VOLLEY accommodation. A real host trade must obtain the stage mass and inertia, permitted envelope, bolt pattern, power/command provisions, recoil and pointing limits, disposal requirements

TABLE V

SERVICE TIERS FOR SMALL-SATELLITE ORBIT PLACEMENT. VOLLEY'S SYSTEM MASS IS 126.6 KG DRY AND 174.6 KG LOADED WITH TWELVE 3U SATELLITES

| Attribute | Spring deployer | VOLLEY (this work) | Transfer vehicle |
|---|---|---|---|
| Release / transfer | low-m/s device-dependent | 16.029 m/s model point | mission-dependent |
| State targeting | impulse and host-dependent | 0.0274 m/s simulated $3\sigma$ | service-dependent |
| Payload interface | qualified per device | compatibility unverified | service-dependent |
| System mass, 12×3U | ∼72 kg comparator | 126.6 kg dry model | quotation needed |
| Plane/LTAN change | host/drag-dependent | not demonstrated | service-dependent |
| Evidence status | flight heritage | computational study | provider-specific |

and provider safety process. The parametric recoil cases in Table IV must not be read as compatibility findings.

### C. Integration Interfaces

A candidate integration would need a provider-approved mechanical interface, an electrical recharge and inhibit interface, an authenticated command/telemetry interface, and verified attitude, thermal, magnetic, debris and disposal constraints. The current CAD includes only a reference ring shape and bracket assumption; it does not establish ESPA or ESPA-Grande accommodation. An assumed 150–300 W recharge allocation, state-vector input and shot authorization are model inputs rather than approved host services. The deployer would command the modeled release only inside a host-authorized window, and the host would retain attitude and disposal authority. Payload-side exposure also matters: the modeled permanent-magnet field at the near payload face exceeds a representative magnetometer full scale, so compatibility requires a named payload and an explicit exclusion, standoff or shielding decision (Sec. XII).

A multi-shell mission could combine host manoeuvres and deployer release impulses, but its value must be calculated against spring release, differential drag, onboard propulsion and orbital transport with the same manifest and deadlines. A prior claim that release timing alone created persistent phase has been withdrawn. No complete twelve-payload campaign closes in the six sampled finite-burn cases, so this paper does not claim a delivered four-shell or twelve-target service.

### D. Position in the Deployment Landscape

Spring deployers provide a mature reference for low-velocity separation; a host manoeuvre, differential drag or an orbital transfer service can add mission-level placement. VOLLEY is an unbuilt computational candidate for commanded release. Table V contrasts the studied functions and flags where a common-mission comparison or supplier-backed mass is absent. The approximate 6 kg spring canister used elsewhere in this paper implies about 72 kg for twelve 3U units, while the modeled Gen5 dry mass is 126.6 kg. No gap in the commercial market is established by this technical comparison.

### E. Demonstration Path

### F. What Air Costs a Ground Test

The proposed orbital machine would operate in vacuum; the velocities above are computed without aerodynamic drag. A

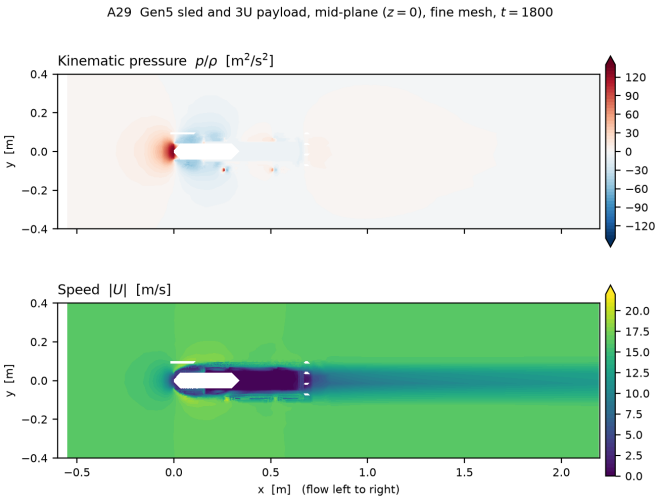



Fig. 18. Mid-plane slice ($z = 0$) of the fine-mesh CFD screen: kinematic pressure above, speed below, flow left to right. Stagnation on the forward face, separation at the shoulders, and a wake that has not recovered by $x = 2.2$ m. The pressure term shown here is integrated directly from the solved field; the viscous term is not solved and is bounded by a turbulent flat-plate correlation over the meshed wetted area, as the text states.

future ground test might fire a mass simulator in air, but no VOLLEY ground shot or measured velocity exists. The CFD case below is a pre-test sensitivity at a historical 16.388 m/s model point, not a correction validated for the present finite-force shot. A steady RANS solution ($k$-$\omega$ SST) around the sled and payload *as generated*, meshed directly from the CAD surfaces rather than from an idealised body, gives a drag coefficient of 0.52 on a 201.6 cm$^2$ projected frontal area at $Re = 7.0 \times 10^5$, i.e. 1.73 N at exit velocity. Because the profile is position-scheduled, $v^2$ rises linearly with distance and so does the drag, making the work over the 1.30 m acceleration zone 1.13 J and the exit-velocity deficit 5.1 mm/s, or 0.031 % (Fig. 20).

That figure is small against the design point and *not* small against the quantity a ground test resolves: it is 19 % of the $3\sigma$ closed-loop dispersion. An atmospheric ground test can be considered for initial velocity measurements, but its air-load correction and uncertainty must be reported; vacuum and thermal testing remain separate qualification questions. The distinction depends on how the test is run: open-loop, the deficit appears as 5.1 mm/s of velocity; closed-loop, the servo nulls it and it appears instead as 1.73 N of additional commanded force, 0.125 % of the shot.

Two properties of the solution are reported rather than presented as convergence. The velocity residuals fall three orders and then plateau near $5 \times 10^{-2}$, which is what a steady solver does on a massively separated wake, so the force is a mean over a window of iterations quoted with its spread, $1.734 \pm 0.144$ N. And the viscous contribution is a turbulent flat-plate bound over the meshed wetted area rather than a solved wall shear stress, 17.7 % of the total; the pressure term, which dominates, is integrated directly from the solved field

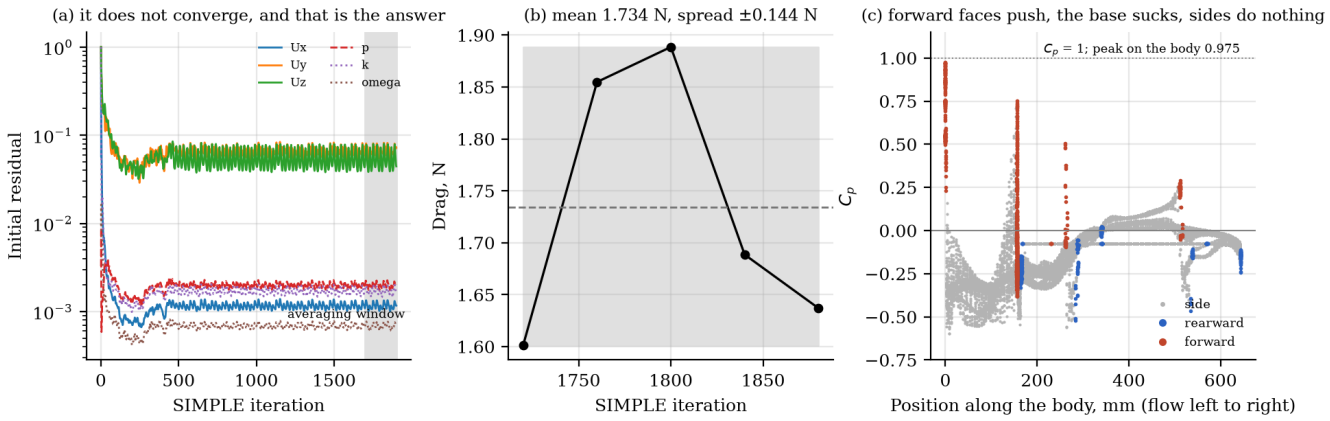



Fig. 19. Force history and mesh convergence for the same case. The force is a mean over a window of iterations quoted with its spread, $1.734 \pm 0.144$ N, because a steady solver on a massively separated wake plateaus rather than converges; $6.3\times$ the cells moves the answer by 4.86 %.

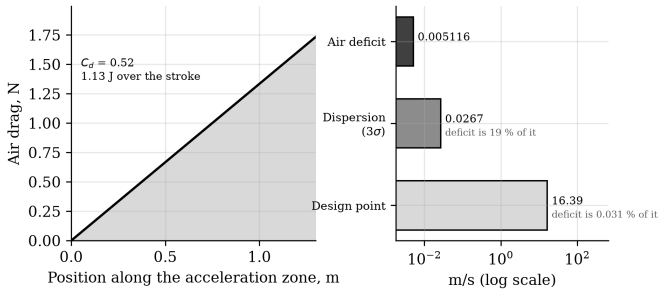



Fig. 20. Air drag along the acceleration zone, and the resulting exit-velocity deficit set against the historical 16.388 m/s model point and a modeled dispersion a future ground test would need to resolve; no shot has been measured.

over the body surface. Adding the stator plates *reduces* drag by 13 % and halves the oscillation, the plate acting as a wake splitter rather than as a confining wall, so the free-stream figure quoted above is the conservative one.

A future instrumented ground article could measure sled and payload velocity, feed forces, tip-off, brake temperature and cycle reliability against predeclared bands. Its length, containment, cost and site approvals require engineering and a quotation; this study supplies neither. A captive hosted test would require a named platform, provider interface and safety approval even if the payload were never released.

## VII. Sensitivity and Uncertainty

Because several inputs carry manufacturing or environmental spread, the historical shot point is bracketed by a first-order sensitivity analysis (Table VI). Each coefficient is the local partial derivative of exit velocity or thrust with respect to the listed variable, evaluated at that historical point. The dominant contributors are payload mass and magnet remanence; the velocity model is mild in all of them, and the closed-loop servo further collapses the delivered dispersion to 0.0274 m/s (Sec. V-A) because it commands current against measured position rather than trusting the plant. Root-sum-

TABLE VI
First-order sensitivities of the historical shot model

| Variable | Affects | Sensitivity |
|---|---|---|
| Payload mass | $v$ | $-\frac{1}{2}$ (%/% of $m_{\mathrm{tot}}$) |
| Magnet remanence $B_r$ | $F, v$ | $+1.0,\ +0.5$ %/% |
| Current density | $F, v$ | $+1.0,\ +0.5$ %/% |
| Winding resistance | $\eta$ | $-0.15$ %/% |
| Capacitor ESR | sag | $+0.04$ V/m$\Omega$ |
| Gap asymmetry | $K_t$ | $-13$ %/mm |
| Sensor noise (8 mm/s) | dispersion | sets $3\sigma$ floor |
| Temperature ($B_r$) | $K_t$ | $-0.11$ %/K |

square combination of the manufacturing terms (magnet grade $\pm 2\%$, gap $\pm 0.05$ mm, resistance $\pm 5\%$, ESR $\pm 20\%$) gives an open-loop velocity spread of $\pm 2.1\%$, consistent with the Monte Carlo of Fig. 10. Environmental terms are handled by rating margin: the N45SH remanence temperature coefficient of $-0.11\%$/K over a $\pm 40$ K operating band moves the thrust constant by $\mp 4.4\%$, so the worst-case cold-to-hot requirement of 132 kA/m stays inside the 140 kA/m current rating without re-tuning. Atmospheric and solar-activity spread do not affect the machine and enter only through orbital lifetime, where Sec. V-B records that the sweep previously offered as evidence of invariance could not, by construction, have detected a change.

## VIII. Mechanical Analysis

The structural load path is sized to the two governing cases: launch vibration during ascent and the arrest impulse during operation. Fastener and pin margins are quoted on ultimate strength at a 1.4 design factor per common launch-hardware practice, consistent with the qualification philosophy of GEVS [21] and ECSS structural standards [22].

*Track stiffness.* A provisional structural screen seeks a first natural frequency above a cited 70–100 Hz secondary-structure range; the applicable provider requirement is unknown. Modeling the twin-longeron track as a uniform beam carrying the 20 kg stator distributed mass, the fundamental

is $f_1 = (\lambda_1^2/2\pi L^2)\sqrt{EI/\mu}$ with $EI$ from two 7075 box sections. A simply-supported idealization gives 48 Hz, but bracketing both ends into the ESPA ring (fixed-fixed) raises the idealized first mode to 109 Hz. This assumed boundary condition and an intermediate rib are design-study inputs; no selected host interface or provider-specific modal acceptance has been demonstrated.

*Inter-array attraction.* The two Halbach faces attract across the gap with a Maxwell stress $p = B^2/2\mu_0 \approx 120$ kPa at the 0.55 T mean working field, or 3.7 kN over the array footprint. Carried as a bending load into the titanium side plates, this produces 33 MPa against the 880 MPa Ti-6Al-4V yield, a margin above 20; this idealized static plate calculation has a margin above 20 against yield, but it does not validate the as-drawn assembly, joints or launch-load path. The flat-plate expression overstates the force: a three-dimensional integration of the field gradient over the meshed blocks returns 2.69 kN, some 37% lower, because the assumed 0.55 T is not the mean normal field over the separation plane. Jensen's inequality acts in the opposite direction and does not explain the overestimate. Extending the Maxwell-stress plane beyond the magnet footprint gives 2.680 kN, within 0.25% of the 2.687 kN volume-force calculation; this is model-to-model agreement, not experimental validation. The earlier finite-element load remains conservative.

*Arrest and roller loads.* At the 200 $g$ arrest cap the 9.445 kg sled develops 18.5 kN axially, reacted by the ring-spring stop; the pitch couple during braking loads the guide rollers to roughly 1.48 kN per pair. This is the load path most affected by adopting the CAD-derived sled mass, it very nearly doubles, and it is the one place where the heavier sled makes the machine harder rather than merely slower. A 1.48 kN per-pair roller load requires a source-checked bearing, mounting and lubrication design; reuse duty is not addressed here: the sled is braked twelve times per campaign in vacuum, and no lubrication or contact-life assessment has been performed. The abort latch, engaged only for a commanded abort, carries the 13.4 kg carriage at the abort deceleration, 0.42 kN, through dual A-286 pins at a margin above eight.

*Fatigue and tolerance.* A static bond-stress screen gives 0.12 MPa at the assumed 200 $g$ arrest case against a provisional 10 MPa allowable. It does not establish bond fatigue, shock, thermal-cycle or debris performance over a service life. Positional accuracy is set by the winding-to-magnet gap, whose $-13\%$/mm thrust sensitivity dictates a $\pm 0.05$ mm shim tolerance to hold thrust within $\pm 0.65\%$; this is a proposed precision-shim target, without a released tolerance stack or demonstrated manufacturing capability.

### A. Architecture reliability, and the bar it sets

A spring dispenser is twelve independent one-shot mechanisms in parallel: one failure costs one satellite. This design is one mechanism in series with itself, cycled twelve times. The sled, stator, converter, energy store, sequencer and brake serve every shot, and the escapement and retention gate cycle twelve times each, so a failure at shot $k$ forfeits shots $k$ through twelve. Enumerating the shot chain, nine of thirteen elements forfeit the remaining manifest on a single failure, against zero for a spring dispenser, and nine shared elements over twelve cycles is 108 opportunities to fail. This is structural and no component quality removes it.

What component quality does change is where the comparison lands. Writing $p$ for the probability that the shared chain survives one shot, expected delivery is $\sum_{k=1}^{12} p^k$, which departs from a spring's independent $12q$ quickly. Against a spring at $q = 0.99$, matching on *satellite count* requires $p = 0.9985$, which is not a realistic target for a twelve-cycle electromechanical system without flight heritage. Matching on *delivered orbital life* requires only $p = 0.9347$, because each delivered satellite is worth 1.50 times a spring-deployed one. Expressed per element per cycle, the requirement is $r \geq 0.99326$.

That is the first quantitative reliability requirement this design has carried, and it is not yet met or measured: no cycle-life test exists for the escapement, the gate or the sled. Two candidate mitigations are defined but are deliberately not credited above, because crediting them without analysis is the optimism the exercise exists to avoid. Retention gates are per cassette, so one gate fault forfeits six rather than twelve. And the winding is segmented: a dead segment is a length the sled coasts over rather than a stopped machine, so at four segments a failure downstream of the breech still exits at 14.19 m/s, 86.6% of nominal. The exception is the breech segment itself, where no force acts on a stationary sled and the shot never starts, which makes duplicating or overlapping that one segment the cheapest available reliability action.

## IX. Thermal Analysis

The shot model assigns approximately 855 J to stator copper, 1162 J to sled braking, 93 J to the converter and 78 J to the bank ESR per rated cycle. Multiplying an assumed twelve-shot total of 26.6 kJ by a uniform 13.5 kJ/K structure thermal capacitance gives a 2.0 K *bulk-average* rise; it says little about local peak temperatures. The corrected 0.344 kg fin mass gives a 9.0 K adiabatic rise per shot and a 108 K twelve-shot no-cooling bound. A transient conduction/radiation model has not established cooling between shots at the adopted 1200 s cadence. The 0.32 m$^2$ radiator estimate uses an assumed 130 W rejection case and is not a closed thermal design. Component survival temperatures, heater power and irreversible magnet loss require supplier data and orbit-specific hot/cold cases. The historical-case thermal outputs remain model estimates, not environmental qualification.

## X. Power Electronics

The electrical architecture proposes a 96 V, thirty-two-cell pulse bank and a three-phase SiC inverter with position-synchronized control. The winding model predicts 19.7 $\mu$H, a 373 A peak phase current and 25.2 m$\Omega$ per phase at the historical modeled point; 320 A is the reported DC-link peak and is a different quantity. At the assumed 20–40 kHz switching range, the calculated current ripple and copper contribution are

TABLE VII
PRINCIPAL FAILURE MODES AND MITIGATIONS

| Failure | Effect | Mitigation |
|---|---|---|
| Sled jam | Manifest stranded | Caged escapement; controlled-disposal host. The dual-cassette split bounds an *escapement or gate* fault to half the manifest, but not a sled fault: one sled serves all twelve cells |
| Latch fails | Early release | Latch loaded only on abort; inertia retains during accel; three-inhibit fire chain |
| Sensor fault | Wrong velocity | Redundant seat switches, encoder, photogate cross-check; no-fire on disagreement |
| Capacitor cell | Reduced energy | 32 series cells; one shorted cell drops rail 3%, within margin |
| Inverter device | Phase loss | Segmented winding; dual gate drive; degraded-thrust continue |
| Magnet debond | Field loss | 84× bond margin at 200 $g$; taper-limited arrest |
| CubeSat misalign | Feed jam / tip-off | Seat switches + optical check; no-fire interlock |
| Brake fault | Sled overrun | Ring-spring hard stop backs eddy brake; motor regen trim |
| Controller fault | Loss of shot | Safe-state inhibits; watchdog; abort to 45% stroke |

model outputs. Device ratings, switching overshoot, gate-drive loss, cell balancing, fault-clearing energy and flight derating have not been checked with a selected part list or a physical converter.

Current limiting, bus clamping, desaturation shutdown, balancing, bleed/passivation, segmentation, shielding and dual sensing are proposed controls. None has a closed circuit schematic, thermal qualification, EMC measurement, fault-injection result or provider acceptance in this study. A shorted segment can halt a shot or create an unsafe transient; segmentation cannot yet be credited as fail-operational. The permanent-magnet field and inverter transients may reach other payloads, so scheduling after primary separation is not a substitute for payload-specific magnetic and EMC analysis.

## XI. Reliability and Failure Modes

Table VII summarizes the principal failure modes in an FMEA form, with the mitigation designed into the architecture. The intended safety goals are to retain the satellite through ascent, prevent unintended release after any credible fault and recover a safe state when possible. Single-fault tolerance and graceful degradation have not been established for the feeder, bank, inverter or brake.

## XII. Space Environment, EMC, and Standards

LEO accommodation would require a selected material and component list checked for outgassing, atomic oxygen, thermal cycling, radiation, vibration and shock against the actual provider and mission standards [22]. The current model proposes low-outgassing polymers, protected surfaces and N45SH magnets, but no supplier qualification record, launch-load test or environmental test is supplied. The calculated 109 Hz first mode and selected load margins apply to stated structural idealizations; they do not demonstrate assembly survivability.

The permanent-magnet sled is a payload compatibility concern. The modelled back-face field is 22.7 mT at 10 mm, 4.3 mT at 20 mm and 0.4 mT at 50 mm for the stated geometry. Across a 3U payload envelope, the calculated field is 61.1 mT at the near face and 0.341 mT at the far face. The near value exceeds a representative $\pm 100$ $\mu$T magnetometer full scale by 611 times; even the far value exceeds it by 3.4 times. These are model results, not a qualified payload field map. No published VOLLEY ICD defines a keep-out radius, and the proposed steel septa and shielding have not been evaluated as installed hardware. Magnetometer recovery, magnetorquer operation, permanent magnetisation of payload parts and switching-transient coupling must be assessed with a named satellite. Timing a firing after primary separation does not remove the static-field exposure of stored spacecraft.

## XIII. Manufacturing, Cost, and Scalability

A possible fabrication sequence uses a wound copper stator on an insulating former, a restrained fixture for assembly of the magnet arrays, and late loading of spacecraft after mechanism inspection. Proposed slot and gap tolerances of $\pm 0.1$ and $\pm 0.05$ mm are design targets, not supplier capabilities or a completed stack-up. The current STEP shapes omit fasteners, harness, joints and feed motion, and the side-fed reference assembly clashes; therefore no released manufacturing drawing set, assembly process or recurring maintenance interval can be claimed. An eventual ground article would require measured current-to-velocity calibration and acceptance of the complete release cycle.

The cost model is parametric and has no supplier quotations. In its assumed bill of materials, avionics, pulse storage and sensing dominate the estimated recurring hardware cost more than the magnet set; this ordering is conditional on unquoted prices and the incomplete installed system. The historical modeled shot draws 2.78 kJ, less than 1 Wh of electrical energy, but that is not a service-cost comparison: host power, storage, integration, qualification, mass, reliability, operations and any host propellant must be counted. A matched mission comparison with qualified spring systems, host manoeuvres, onboard propulsion and transport services remains open.

Ideal scaling trends ($v \propto \sqrt{L}$ at fixed acceleration) motivate a wider trade, but the payload-family table is model output, not accommodation of other CubeSat sizes. More payloads change feeder geometry, structure, reliability and host burden; the current twelve-slot reference already fails a width check. Lunar or deep-space use would require new interfaces, thermal/radiation design, mission dynamics and payload qualification, and is outside this study.

## XIV. Operational Concept and Maturation

The proposed operations sequence is launch with the deployer dormant; host and payload health checks; bank charge inside a provider-authorized window; one commanded release followed by independent confirmation of separation and safe

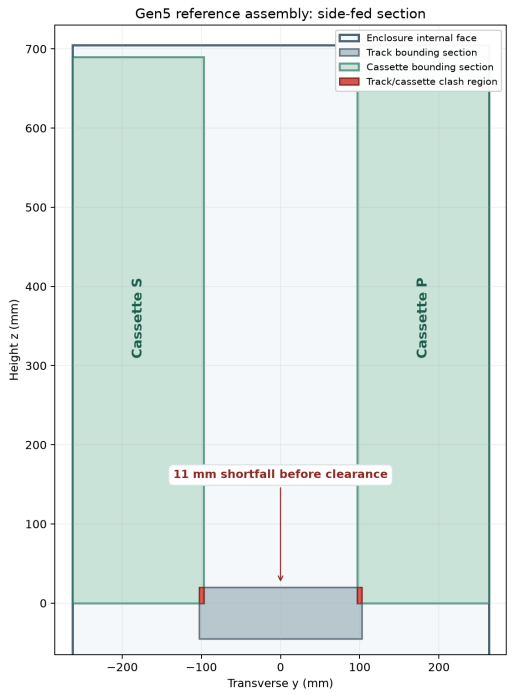



Fig. 21. Gen5 side-fed reference-section failure. The two cassette widths and track require 537 mm within a 526 mm internal width before clearance. FreeCAD and CadQuery exact-solid intersections each find 32,915 $mm^3$ overlap per side. A new feed or enclosure configuration would require renewed analysis.

sled arrest; and repetition only after fault review. Inhibit logic, an early-stroke abort window, bank passivation and eventual host disposal are design requirements, not demonstrated functions. POEM-3 provides a hosted-stage precedent [14], but no POEM or other provider has approved this sequence for VOLLEY.

The next evidence sequence is a reviewed feeder redesign, component-level bench tests, an instrumented integrated ground article, environmental tests against a named payload and host, and only then a provider-approved captive or deployment demonstration. Each stage needs predeclared acceptance bands and a decision review. No technology-readiness level is assigned from the present numerical results, and a future test does not automatically confer one.

## XV. Finite Geometry and Matched Reference Findings

The rated 16.029 m/s shot scales a depth-resolved periodic thrust constant across the 1.3 m powered stroke. The Gen5 CAD instead places a 340 mm mover magnet envelope at $x = 0.230$–$0.570$ m, while its 162 seven-millimetre copper belts on eight-millimetre pitch end at $x = 1.296$ m. The field model represents seven 48 mm wavelengths, or 336 mm of magnets. At 1.066 m sled travel there is no direct geometric overlap between the represented array and the stator. This 4 mm length mismatch and the 1 mm belt gaps are model/configuration discrepancies, not accepted correction factors.

A new screen integrates $\mathbf{J} \times \mathbf{B}$ over the volume of each finite belt using the existing three-dimensional cuboid magnet field, optimizing phase independently at every sled station. This is independent of the periodic-force scaling, but shares the earlier analytic magnet field and is *not* an independent three-dimensional finite-element solve. At the declared rated current it returns 1.042 kJ of ideal work and $\sqrt{2W/13.445\,\mathrm{kg}} = 12.448$ m/s. It omits switching, voltage and friction, so this value is an optimistic geometry-only result within that model, not a validated achievable speed. Cross-section quadrature refinement changes the three checked station forces by at most 0.369%, and halving the 25 mm station spacing changes work by 0.096%. A 2-D finite-element vector-potential solve of the finite seven-wave array, integrated through the same 162 belts, gives 1.082 kJ ideal in-plane work on a 1 mm mesh; the 2-to-1 mm mesh work change is 0.59% and an enlarged air boundary changes work by less than 0.001%. It confirms the end-of-stator decline by an independent field method but omits the 90 mm depth end effect. A separate time integration of the analytic finite-force profile with the historical assumed 96 V, 6 F, 12 mΩ bank gives 12.448 m/s and 2.099 kJ gross draw with the full 1.3 m winding energized. A hypothetical 0.34 m active-copper branch draws 1.394 kJ at the same ideal speed. Neither branch selects a real winding or inverter; it omits switching and release contact. A second, depth-resolved 3-D surface-charge implementation integrates the finite magnet faces and all 162 belts without calling the first cuboid-field code. It returns 1.042 kJ, agreeing with the first formulation to $4.35\times10^{-8}$% in ideal work. Face quadrature refinement from 16 to 24 nodes changes work by 0.00002%, and halving the 25 mm station spacing changes it by 0.096%. The two 3-D implementations share geometry, uniform remanence and ideal phase, so this is numerical formulation evidence rather than a 3-D FEM or independent physical-model validation. An illustrative 14 mm face gap gives 0.907 kJ, 13.0% below the 12 mm nominal case; a 16 mm gap gives 0.790 kJ. A 10 mm copper-depth offset gives 0.961 kJ. These deterministic scenarios are not measured manufacturing tolerances or statistical uncertainty (Fig. 23). Figure 22 compares force methods.

A further quasi-steady three-phase voltage/current screen retains the unselected historical winding. The 12 mΩ reference source does not bind voltage and returns 12.448 m/s; half the assumed current limit gives 8.802 m/s. With the older 116–185 mΩ distributor-data range for a single commercial capacitor string, source power fails at 0.612 and 0.160 m respectively. Ideal two- and three-parallel-string branches complete numerically while multiplying cell count, capacitance and source mass. They are not manufacturer-rated choices. The model omits switched-current transients, heat and contact (P125). R1's nominal 5 mm side clearance leaves 1.31 mm under an assumed 0.2° alignment stack but overlaps by 0.47 mm at 0.5° (P126); these are study bands, not released tolerances. At the ideal 12.448 m/s speed, the modeled 9.445 kg sled carries 732 J, requiring at least 37.6 $g$ mean deceleration across the provisional 210 mm brake corridor (P127). Neither peak arrest load nor host attitude is verified.

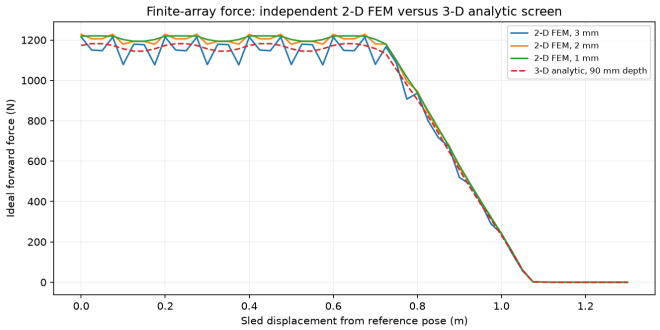



Fig. 22. Finite Gen5 force profiles from a 2-D magnetostatic FEM at three mesh sizes and the 3-D analytic cuboid-field screen. Both use ideal phase. The 2-D solve omits magnet-depth effects; no curve is a measurement.

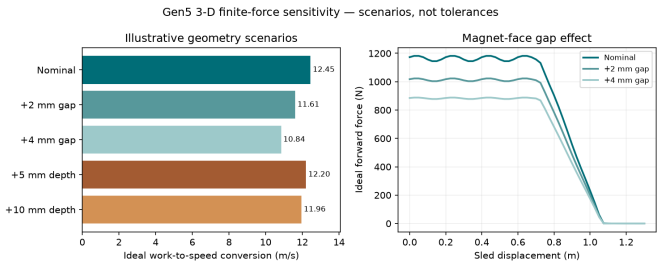



Fig. 23. Deterministic ideal-force scenarios for face gap and depth offset. The 14 mm face gap gives 0.907 kJ versus 1.042 kJ at 12 mm; these are assumed geometry changes, not measured tolerances or achieved speed.

The as-evaluated side-fed assembly still fails its fit check. A separate unselected R1 FreeCAD/CadQuery geometry experiment widens the external enclosure from 530 to 570 mm and exports a native document, STEP parts and two assembly STEP files. Exact-solid static checks report zero track–cassette overlap; twelve sampled 3U-envelope paths and their axis-aligned swept boxes clear listed fixed parts. The candidate has no lift carriage, actuator, launch retention, tolerances, supplier mass or provider envelope approval. Its 0.99 kg enclosure-skin increment is a simple material-volume proxy, not a revised installed mass. It must not be combined with the old shot and mass figures as if it were an accepted Gen5 configuration.

A common-input mission screen compares twelve 4 kg payloads at 450 km on a 300 kg base host with 10 kg fuel, 2 kg reserve and assumed 220 s host specific impulse. A 72 kg spring-dispenser class and 126.562 kg modeled Gen5 installed dry mass are included in host mass. For one same-epoch target of $+16.029$ m/s inertial tangential payload velocity, an ideal host preburn and momentum-conserving recoil require 2.693 kg of host propellant for a 2.5 m/s spring release, 0.827 kg for the finite-force 12.448 m/s geometry-only case and 0.030 kg for the contradicted historical rated case. The spring mass is a class assumption rather than a vendor quote; the Gen5 mass includes assumed items.

With the same 20-minute cadence, 5 km arc-spacing targets, 18,000 s terminal epoch and assumed 100 N host thrust, the existing finite-burn optimizer finds accepted prefixes of

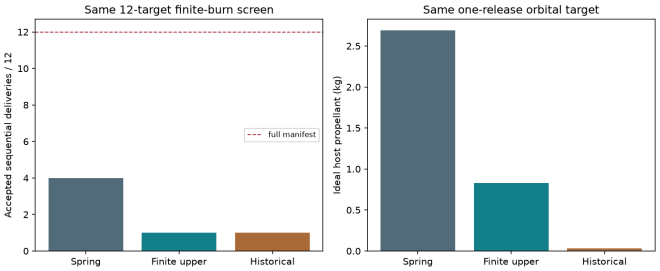



Fig. 24. Matched reference inputs and bounded optimizer outcomes. No sampled twelve-shot option closes; assumed device masses and host performance are not supplier data.

4/12, 1/12 and 1/12 respectively. Its 64-corner replay accepts the spring prefix in 24 corners and neither Gen5 prefix in any corner. This is a bounded optimizer result, not proof of mission impossibility or commercial advantage. Figure 24 and the local run records give exact inputs and traces. A named payload/provider and matched orbital-transport, onboard-propulsion and differential-drag alternatives remain open.

### A. *Delivery-value and prototype requirement screens*

The matched one-event reference was also re-solved with each option's initial fuel set to that event's ideal burn plus an equal 2 kg reserve. Relative to the 72 kg spring class, the 126.562 kg Gen5 device saves 1.866 kg of fuel when both initially carry 10 kg, but its dry device is 54.562 kg heavier. With fuel resized for one event, device plus fuel remains 52.717 kg heavier. At the 12.448 m/s ideal finite-force speed, device mass would need to fall to about 73.932 kg for parity on this narrow proxy. Even a perfect zero-burn release cannot exceed 74.659 kg of device and tie that one-event reference with the common reserve. These are necessary architecture screens, not full-campaign economic conclusions; a customer may value a state and deadline that mass alone does not price.

To turn that proposition into a testable requirement, an independent ideal two-body sensitivity screen maps specified 450 km circular-orbit semimajor-axis targets to tangential inertial velocity. A 20 km axis rise requires 11.149 m/s, while a $\pm 0.1$ km axis band leaves only about 0.0555 m/s on its smaller side if the *entire* error were assigned to tangential velocity. Host state, pointing, timing and tracking would consume part of that budget, and semimajor axis alone does not define precision delivery. Ideal constant acceleration over Gen5's 1.3 m stroke permits 15.97 m/s at 10 $g$; 1 km/s over that stroke requires about 39,220 $g$ and 2 MJ of payload kinetic energy for a 4 kg 3U. Consequently the first Gen6 prototype should derive a lower-speed, payload-specific command envelope from an actual delivery requirement. The 1 km/s objective is a separate, longer-stroke or different-payload research problem. Recomputable inputs and sensitivity outputs are published with the companion package; none demonstrates release accuracy or a selected prototype.

## XVI. Limitations

The 20-instance FreeCAD reference assembly is reviewable as native FCStd and STEP, but its as-drawn side-fed placement fails the 11 mm width screen and has exact track–cassette overlap (Fig. 21). An all-leaf exact-solid audit identifies 17 intersecting pairs, including sled–track (82,983 $mm^3$) and sled–stator (6,300 $mm^3$). The widened R1 candidate removes the cassette–track clash but retains five intersecting pairs. Interface-ring and stored-payload proxy overlaps also require physical interpretation and disposition. The scripted feeder path therefore cannot be used as evidence of a collision-free assembly. It is not a solved feeder, tolerance stack or manufacturing release. Eight source solids are CadQuery-generated and re-exported through FreeCAD; their native FreeCAD instances do not carry parametric feature histories. Seven limits bound the claims. The first is that payload structural compatibility is not established, and cannot be established by this work alone. The 25 $g$ ceiling this machine is designed to is a requirement it sets on itself. It was previously presented as following from the CubeSat qualification environment, by integrating a GEVS random-vibration spectrum to 14.1 $g_{\mathrm{rms}}$ and taking three times that as a bounding load; that reasoning is withdrawn, because a root-mean-square level over a broadband random spectrum is not a quasi-static limit load and a $3\sigma$ multiple of it is a peak-response estimate for that spectrum rather than a structural capability. No universal quasi-static qualification level for CubeSats exists to replace it: the CubeSat Design Specification [23] defers test levels to the launch provider, so they vary by vehicle and mission. The defensible statement is therefore narrow: the historical modeled acceleration is 10.1 $g$ for 162 ms, and whether any specific satellite tolerates it depends on that satellite's own qualified load environment and on an integration review that has not been performed. No payload has been mechanically qualified against this machine, and the payload-family accelerations in Table III, which run to 14.0 $g$ at PocketQube, carry the same caveat. The atmosphere model is static, so absolute lifetimes carry severalfold uncertainty; an earlier version of this paper offered the invariance of the lifetime *ratio* as the defensible result in their place, and Sec. V-B records why that was wrong, the sweep supporting it could not have detected a change. The ratio is now quoted at a stated activity level and carries the same uncertainty as everything else derived from a static atmosphere. The winding model has a depth-resolved analytic thrust constant, but the finite-stator force screen in Sec. XV reveals a much larger end and geometry effect; independent finite-element force closure remains open. The brake law is first-order plate drag and remains an unverified arrest approximation. Conjunction figures, although refined at candidate minima, do not replace per-shot collision-avoidance products. Host budgets are parametric wherever stage properties are not public. The 9.445 kg sled input comes from historical Gen3 CAD solid volumes, whereas other dry-mass lines are modeled or assumed. This is not a mass extraction from a complete Gen5 assembly. The former 4.86 kg parametric sled estimate disagrees with those volumes. The CAD-derived value is adopted here and sets the headline velocity, but it is the as-drawn, unpocketed geometry. The historical idealized chassis-plate analysis reports a 17-fold modeled stress ratio; this does not establish an assembly strength margin or demonstrate that a ribbed redesign would recover mass. The sensitivity, mechanical, thermal, power-electronics, and reliability analyses added here (Secs. VII–XIV) address the principal engineering questions only at analysis level. Several remain feasibility-significant. The commercial capacitor strings examined do not meet the ESR requirement; an unselected flywheel motor-generator proxy is modeled at 35 m$\Omega$ against a 68 m$\Omega$ assumption, at modeled mass parity rather than saving; this is not a source-rated hardware result. The three-dimensional depth correction has been applied to the analytic field model and the result was unfavourable: resolving the field across the array's 90 mm depth, rather than sampling it at the centre plane and multiplying, reduces $K_t$ by 4.42% to 10.54 N per kA/m and $v_{\mathrm{exit}}$ to 16.03 m/s. The depth-resolved periodic model is retained for historical comparison; the finite-geometry screens determine the later adverse force finding. A separate three-dimensional solve supports the underlying field model, a reduced-scalar-potential magnetostatic PDE on a 315,370-node tetrahedral mesh agrees with it to 0.059% at midgap, but it tests the field, not the depth average, and no independent three-dimensional finite-element check of the integrated thrust constant itself exists. The earlier 11.03 N per kA/m and 16.39 m/s appear here only where a superseded value is named as such. Hardware validation has not begun.

### A. Numerical evidence reconciliation

A separate orbit-averaged DOP853 integration of specific-energy and eccentricity-vector rates reproduces the static-atmosphere lifetime calculation without importing its historical stepping routine. At 450 km and 61 kg/$m^2$, the no-impulse lifetime is 1.303955 yr; imposed inertial increments of 2.5, 12.448437 and 16.029 m/s give ratios 1.082110, 1.451222 and 1.602165. Refining anomaly quadrature from 360 to 720 nodes and relative tolerance from $10^{-7}$ to $10^{-9}$ changes a ratio by at most 0.000145%, below a 0.5% numerical band. This is agreement under a shared static atmosphere, not a space-weather, achieved-speed or environmental validation.

The corrected A15 R1 GMAT R2026a archive was reparsed from twelve synchronized 59,590-row reports. Its final shortest circular RAAN span is 14.558545°; the old 367° interpretation is withdrawn. A focused five-second window gives 190.149944 m minimum sampled separation. Sparse outputs do not establish continuous clearance. The host remains fixed in the preloaded initial-state construction, so this is not a coupled twelve-release mission.

The full 23-line historical mass rollup is 126.561588 kg; its axial lump CG is 0.457690 m. It is not an installed BOM or complete inertia tensor. Every one of the 17 Gen5 and five widened-R1 solid intersections is retained as a conflict or unresolved interface/proxy intersection. Historical isolated structural cases cannot qualify this assembly. The legacy

A1 field screen used scikit-fem, not FEMM, and has two missed field bands. The pinned Chrono, Basilisk, Orekit, native FEMM, Elmer and selected-source ngspice cases remain unrun. These limitations prevent a hardware performance or integration claim.

## XVII. Conclusion

Gen5 is a fixed, system-level computational study of a magazine-fed linear-motor deployer. Its historical periodic-force model reports a 16.0 m/s 3U shot at 10.1 $g$, 18.8% net electrical-to-payload efficiency and 0.0274 m/s simulated dispersion; the dry mass rollup is 126.6 kg. A finite-array, finite-stator force integral challenges that speed claim, giving 12.448 m/s only under ideal phase. An independent 2-D FEM confirms the force decline but omits depth effects. Under the assumed historical bank and full-winding copper, the finite-force rerun draws 2.099 kJ gross; a single-string source fails at the available ESR bound and switching/hardware ratings remain open. For an assumed historical 16.029 m/s prograde impulse, the orbital model predicts a 28.8 km semi-major-axis increase and a 1.60 lifetime multiplier at mean solar activity; neither is evidence that Gen5 can deliver the input. The commanded release profile is the proposed capability. The model also fails its mass-per-satellite criterion, and its side-fed reference CAD assembly has multiple exact-solid interferences, including cassette–track, sled–track and sled–stator pairs. It has no measured launch, payload qualification, selected host or completed integration review. The result is an analysed design with explicit limits, not a flight-qualified product. The matched reference screen closes no twelve-shot option, and an R1 feeder geometry is unselected. A separately modeled Gen6 one-cell feeder also intersects the first-article guide when the two studies are combined; R1 clears sampled collision states but leaves a handoff gap, while R2 nominally bridges it; neither has checked contact loads or fault-contained release. Independent full-force verification, a selected configuration, a named mission and physical tests are needed before claiming deployment or economic advantage.

The separate FM08-R3 follow-on selects a guided lead-screw feeder at the computational academic-design level (Sec. XVIII-J). Its geometry, sizing and executable sequencing evidence support a defined next design, while measured repeatability, dynamic contact, environmental qualification and host acceptance require later physical work. This positive architecture decision does not change the negative Gen5 system finding.

## XVIII. Gen6 Development Studies and Future Qualification

Gen5 remains the sole evaluated design snapshot reported here; the finite-force discrepancy prevents selection of a performance rating. A separate Gen6 follow-on now specifies a surrogate service study: two passive 1U and two passive 3U spacecraft released within one orbit from an approximately 500 km sun-synchronous host, with four distinct position–velocity targets evaluated 24 hours after the final release. In a fixed-time, ideal-impulse J2 construction, both commanded release and spring plus host maneuvers reach those same targets. Under assumed 30 and 16 kg installed devices, commanded release saves 3.2823 kg of host fuel but carries 14 kg extra device mass; this narrow device-plus-fuel comparison is 10.7177 kg worse. A $+0.1$ m/s release-speed perturbation produces roughly 25–27 km position difference after one day. The values are a mathematical mission screen, not measured accuracy, an optimized policy, customer tolerance or product benefit; full inputs and reproducible outputs are in this repository's P130 record.

Two native FreeCAD/STEP first-article envelopes compare a captured permanent-magnet carrier with a conductive-plate carrier and a passive 4 kg 3U dummy. Their partial frame/guide/carrier material masses are 9.926 and 9.945 kg. Under the same surrogate mission, only about 9.35 kg remains for the unmodeled drive, source, brake, restraint, control and mounting before device-plus-consumed-fuel parity is lost. Five nominal travel stations clear, whereas an assumed 0.7° yaw plus 1.5 mm linear stack fails the guide clearance screen. At 6 m/s, the moving-energy floor is approximately 82 J; an assumed 10% electrical-to-kinetic efficiency implies at least 818 J input, not a selected bank or motor rating. The article's proposed 3–6 m/s measurement range also does not cover the surrogate campaign's approximately 7 and 9 m/s commands. Candidate A is an analysis lead and B remains a challenger; neither has a complete circuit, contact/release system, brake or installed bill of materials. These post-study artifacts are locally archived as P130–P132; they do not revise the Gen5 numerical result or authorize construction.

A separate Gen6 four-cell packaging screen places two 1U and two 3U passive envelope proxies in staggered or paired fixed cassettes around one rail. Both variants have 13 nominal FreeCAD/STEP states without represented solid penetration and sampled transfer and stowed-carrier paths. The staggered final cell has 330 mm of modeled drive travel before the brake boundary, demanding 5.56 $g$ ideal mean acceleration for 6 m/s. Pairing equal-size cells opposite one another raises the shortest travel to 900 mm and lowers that ideal mean to 2.04 $g$; neither value is a motor or release rating. The layouts still lack positive restraint, loaded contact, tolerance closure, carrier docking accuracy and a host interface. Source geometry, STEP verification and the same-boundary calculation are included in the companion CAD record.

Gen6 is therefore a proposed first instrumented prototype programme for a mission-derived lower-speed command range. A separate long-range research question is whether individually commanded orbital release can ever extend toward a 1 km/s-class exit velocity and, if justified by mission analysis, beyond. At constant acceleration, 1 km/s requires about 2.0 km of stroke at 25 $g$ or 0.51 km at 100 $g$ from rest. Achieving it within Gen5's 1.3 m acceleration stroke would require approximately $3.9 \times 10^4$ $g$. The payload load limit, guide geometry, energy and heat rejection, recoil, containment, interface and service value must be established together. Earlier gas-guide and spring-cell studies are unselected compar-

isons. The next gates are an actual payload/host requirement, complete installed-system and safety closure, independently checked coupled dynamics and calibrated hardware tests. None is closed by the Gen5 calculations or the surrogate Gen6 screens.

### *A. Supplementary computational follow-up, 19 October 2026*

A separate native GMAT R2026a–Orekit 13.1.6 check harmonizes the JGM2 constants and prescribes GMAT's exported Earth-fixed pole in the comparison basis. A single continuous GMAT arc gives a maximum position difference of approximately 1.285 mm over one day, compared with the earlier unmatched approximately 1.499 km case. Pole configuration explains the dominant difference; repeated stop/restart histories also affect the numerical comparison. Shared prescribed orientation and small solver residuals do not validate physical orbit accuracy or achieved release speed. Raw states, Java source, native scripts, independent Cartesian equations and refinement results are held in the local harmonization supplement.

A new coupled ideal-impulse construction includes four and twelve alternating 1U/3U payloads, finite-host shot momentum, pre-release and recovery impulses, propellant use and final host-state restoration. The 3–6 m/s first-article input reaches the constructed targets with host assistance; this does not rate the motor. The twelve-release spring case exhausts a 10 kg reserve but completes with 20 kg initially available, consuming 15.35 kg; the corresponding VOLLEY case consumes 4.05 kg. With assumed 48/90 kg device proxies, fuel saving does not recover the device-mass penalty. The targets, ideal impulses and proxies are assumptions, not an optimized customer mission. An adaptive interval screen applies only to nominal point centers after a 20 s host guard and a stipulated 1 cm state-error bound; it is not physical launch-interface or flight-conjunction clearance.

The local seven-feeder supplement supplies fourteen native candidate assemblies, STEP configurations, analytical force/torque/tolerance records and a six-case native CalculiX floor-fixture screen. Connected fork geometry is distinct from verified joint, handoff and power-off restraint behavior. Installed actuator and latch ratings, full-cycle contact, fault containment and equal-boundary four/twelve-cell integration remain incomplete. A rectangular capture-tooth screen reaches approximately 1.97 GPa nominal bending under the largest calculated mean arrest load, exceeding even a 1000 MPa assumed yield band; peak contact, actual material and engagement remain unknown. No candidate passed the original all-concept evidence gate in this dated study; the subsequent academic-design downselect is reported in the FM08-R3 section. These results are follow-on research and do not repair or replace the frozen Gen5 result.

### *B. Captive-transfer and limited replenishment follow-up*

The separately versioned CT-01 coupon supplies 22 native/STEP states and continuous interval nonpenetration checks for 20 prescribed rectilinear phase cases across 1U/3U envelopes. Downward gate retraction and separate receiver-lock bands correct nominal crossings, but impose new accommodation and actuator requirements. Conditional arrest sizing retains 226 of 972 cases exceeding a proposed 24 mm total travel allowance, and 148 of 243 local gap sensitivities are nonpositive. Six native isolated thick-root tooth fixtures retain mesh-dependent peak stress; no rated latch, actual catch dynamics or full launcher integration is established. False lock indication, uncontrolled gate closure and missing actual sled custody remain physical counterexamples. This dated coupon study did not select a feeder.

A limited cooperative-cartridge replenishment concept defines datums, capture/retention, power isolation, contamination, operations and faults, with 108 capture-energy and 36 insertion-fit sensitivities. Eighteen fit cases fail. The current fixed-cassette baseline lacks removable-cartridge integration, a servicing interface and verified double custody; operational replenishment is therefore a no-go for this configuration. This is a bounded negative concept finding, not robotic or docking validation. The local companion reports retain assumptions, source and raw results.

A full-text review of the closest stacked-deployer and mixed-size storage studies clarifies their evidence boundaries. Zhao et al.[30] report gravity-compensated laboratory release ranges of 0.67–1.42 m/s for a 1 kg dummy and 0.40–0.85 m/s for 3 kg; their shared transport route remains vulnerable to a stuck payload. The later storage study[29] employs spacecraft housing grooves, guide features and iron patches, so its receiving method does not establish compatibility with unchanged passive spacecraft. Its approximately 1200 N receiving force is simulated, while the approximately 0.20/0.246 s aligned/offset push tests are gravity-assisted. Those results are not a single validated zero-gravity operating point or VOLLEY test evidence. The companion full-text comparison records page references, source hashes, a motor-sizing arithmetic discrepancy and the resulting interface/contact verification requirements.

### *C. Native coupled replay and precision requirements*

CNR-01 replays the complete prescribed host burn/recoil sequence from one initial state in Orekit 13.1.6. Its 51 native cases include nominal four/twelve-payload comparisons, step refinements and deterministic command-error probes. Nominal payload endpoint differences from the source remain below 8 $\mu$m and twelve-case step changes below 17 $\mu$m. This verifies numerical consistency of separate force/bookkeeping implementations; both use DOP853 and share constructed targets and commands. It does not establish physical accuracy or independently optimized controls.

A common +0.1% imposed release-speed bias produces approximately 1.53 km terminal position error in the VOLLEY cases. Conditional photogate and frame-compensation budgets therefore develop requirements rather than certify precision. Frozen-command thrust-time screens also expose host constraints: at 5 N, the twelve-spring plan needs approximately 6135 s before the 3600 s final release, versus 1488 s for the imposed 3–6 m/s case. These totals exclude finite-burn

retargeting and operational scheduling; another competent spring-host policy may improve them. Actual actuator ratings, feedback, installed mass and near-field integration remain open. The companion repository retains all event records, native logs, sensitivity cases and exact source identities.

### *D. Finite-duration host control, 20 October 2026*

FBR-01 replaces ideal host impulses with two targeted constant-inertial-force windows per leg, integrating mass flow at an assumed 220 s specific impulse. Preparation begins at $-900$ s and is counted; each release is followed by a 20 s zero-force guard, with final host recovery by 4200 s. Eighteen targeted schedules are independently replayed in 46 native Orekit runs from one initial state. Shared constructed targets, the fixed-axis J2 force model and the DOP853 integration family limit independence; numerical agreement is not flight accuracy.

For the declared 35%/35% policy, four-release spring and imposed VOLLEY inputs require 13.36/6.50 N peak and 6.056/1.895 kg fuel. Twelve-release inputs require 33.83/18.00 N and 23.122/6.123 kg, respectively: spring exceeds the illustrative 20 N and 20 kg limits. With a common 30 kg reserve and 100 N limit, both twelve-release policies reach their constructed full-state targets. Their 17.460 kg propellant difference remains smaller than the 42 kg device-proxy penalty. Neither universal spring infeasibility nor a net installed-mass advantage follows.

Interpreting the resultant as a single fixed thruster requires up to approximately 9 deg/s mean direction slew in 20 s zero-force gaps. No actual host attitude, opposed-thruster allocation, plume/contact geometry, navigation feedback or motor rating is verified. Fuel-exhausted formal traces remain diagnostic only. The local finite-force supplement retains all controls, mass histories, native logs, deterministic probes and failed screens; it advances translational targeting without closing operational delivery.

### *E. Revised feeder custody and actuation screen*

The separate FM-02 revision supplies a connected receiving sled, a 12 mm underfloor entry frame and positive endpoint crosspins. Twenty-eight native/STEP poses and 26 continuous prescribed rigid-translation phases have no represented penetration. Inserted pins obstruct the tested motions; actual pin axial retention, release actuators, loaded contact, deployment/reset and installed integration remain unverified. This dated FM-02 study did not select among the seven architectures.

A distinct integrated ball-screw ground-drive screen uses quintic motion and catalogue limits [32], [33]. All 972 sampled decisions agree with polynomial extrema and piecewise integrals; maximum relative 801-sample torque and input-energy-proxy discrepancies are $8.58 \times 10^{-8}$ and $1.08 \times 10^{-6}$. This is numerical consistency, not physical correlation. The study retains 115/324 nonpositive fit residuals, 12/54 winding-temperature failures and brake sensitivities with 96 no-stop cases. Later twelve-release coasts offer 92.182 s: a 103.0 s assumed handling sequence fails eleven releases, whereas a 74.5 s hypothesis fits. Neither hypothesis establishes actuator performance or a complete campaign. The local companion supplement contains the source, native CAD/STEP, logs, failure cases and reference sheets; these separate Gen6 studies do not change the evaluated Gen5 configuration.

### *F. Receiver stiffness and contact-model verification, 21 October 2026*

FM06-R1 adds integral transfer tines, longer receiver guides and a corrected backstop junction. Thirty native assembly/STEP states, seventy individual exchanges and 28 prescribed continuous translation checks have no nominal represented penetration. A separate surface audit is necessary: solid validity alone missed an edge-only backstop/rail connection. This nominal correction is not a strength or tolerance approval.

A revision audit identifies a changed crossbar arrangement, so twelve new native CalculiX cases analyze actual FM06 receiver solids alongside forty historical fixtures. With assumed 200 GPa steel and fully clamped bores, the 1,000 N one-dog 3U mean displacement changes from 1.8829 mm for FM02 to 0.05998 mm for current FM06; its final compliance refinement change is below 0.200%. Historical FM03 yields 0.05811 mm and is not used as current stiffness. The result characterizes this component fixture; real pin contact, guide bearing and support flexibility prevent an assembled-stiffness bound. Four tine meshes yield 0.0101446 mm/N mean end compliance with a final 0.0334% change, using assumed 69 GPa aluminium and an isolated clamped root. No material allowable or qualification follows. Of 1,215 deterministic guide-fit combinations, 1,117 have nonpositive residual width; the count is not a hardware failure probability.

The separate FCE12 study evaluates the contact representation using nine disjoint native convex payload cells and orthogonal mechanism material cells. Full original mass and inertia are retained. Internal kg/mm/s scaling agrees with a no-contact SI replay. Fourteen analytical planar-gap controls detect imposed 10 $\mu$m–1 mm overlaps; the largest checked reported-gap error is 4.28 nm. This calibration applies to the stated planar controls, not general contact accuracy. Ten native sequences reach the 26.5 s transfer/capture/withdrawal endpoint; injected jam and stuck-pin cases abort at 11 s.

Independent full-solid checks cover 84 targeted poses. Two residual intersections, at most 0.000124 mm$^3$, admit a non-intersecting pose after a 1 $\mu$m Y translation; original residuals remain recorded. These are discrete geometric-resolution checks. The 250-to-125 $\mu$s final centre-of-mass changes are 1.509 mm for 1U and 1.466 mm for 3U, exceeding the provisional 0.05 mm numerical criterion. Impulsive peaks remain unsuitable as design loads. The capture excludes free separation and complete reset. Source-rated installed actuation, actual loaded joints, approved payload interfaces and multi-cell integration remain acceptance requirements; no feeder downselect follows.

A separate FM07-R5 geometry revision adds captive receiver pins, a hinged exit gate and prescribed empty-carrier

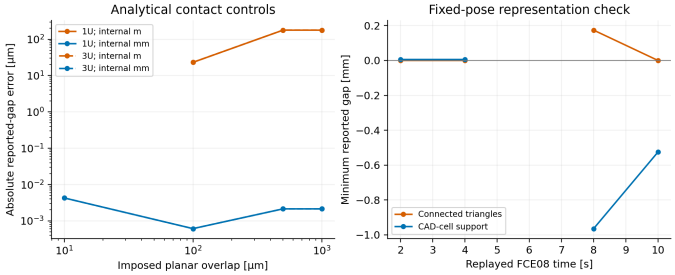



Fig. 25. FCE12 native planar-gap controls and fixed-pose replay. Negative gap denotes detected overlap. The controls establish the stated numerical resolution; final-state convergence and physical contact correlation are separate requirements.

reset. Forty-six native cycle poses, 34 rotary samples and 40 continuous linear sweeps retain no represented nominal collision. Twenty assemblies and 74 individual STEP parts pass independent readback. Rotary clearance remains discretely sampled; loaded fits and installed actuation are separate requirements. The unchanged FM06 fork/tine and receiver-bar material permits component-fixture lineage, not assembled-stiffness acceptance. Modeled single-cell mass proxies are 3.7624/6.1566 kg for 1U/3U, excluding payload and unmodeled installed hardware. The all-seven mandatory selection gates remain unmet; this revision is a development reference, not a selected feeder baseline.

The Japanese original of the Halbach end-effect study [34] uses a 2D infinite-width model, compared with 2D JMAG. Its cancellation conditions require integer magnetic-period length, air-core coils and energized coverage beyond each field end. Selected methods/results passages do not establish finite-width 3D or hardware validation, and no universal correction is applied to VOLLEY. The closely related double-magnetic-circuit CubeSat actuator article [35] remains metadata/abstract-only in this review; no quantitative performance or passive-payload interface claim is inherited.

### *G. Continuous Rotary Clearance and Numerical Contact Diagnosis*

The separate RVC01 study finds zero nominal gate/deck distance near the closed R5 pose, with zero intersection volume. FM07-R6 raises only the panel lower edge by 0.25 mm while retaining its upper edge, hinge and locks. Each gate loses 41 $mm^3$ of material. Adaptive native B-rep distance checks with rigid-rotation displacement bounds cover all four complete opening/closing intervals. The smallest guarded conservative lower bound is 0.0098846 mm. Eight assembly and two gate STEP files pass separate CadQuery/OpenCascade readback; all other components match R5. This is a nominal numerical screen, not a certified kernel error enclosure, tolerance allowance, loaded fit or contact validation.

FCE13 retains the FM06-R1 contact model and physical assumptions and adds six completed native Chrono diagnostic runs. Instrumentation reproduces the original 3U 250 $\mu$s trajectory byte for byte. Sensitivity develops during feed deceleration, before capture. Warm starting, native preconditioned

TABLE VIII
FCE13 POSE REFINEMENT AT THE COMMON 26.5 S ENDPOINT. BOTH RETAINED LIMITS MUST PASS.

| Model / solver | COM [mm] | Angle [deg] |
|---|---|---|
| 1U original PSOR | 1.508584 | 0.639678 |
| 3U original PSOR | 1.466380 | 0.232303 |
| 3U native BB | 1.926456 | 0.120098 |
| 3U exact box / PSOR | 0.716162 | 0.564333 |
| Retained limit | 0.05 | 0.05 |

Barzilai–Borwein (BB) and an exactly equivalent native box-core representation fail to establish pose convergence. The unchanged 250-to-125 $\mu$s endpoint comparisons are: At 125 $\mu$s, substituting the geometrically identical box core alone changes the endpoint by 0.755420 mm and 0.603748°. Work changes below the separate 1% screen do not imply pose or force convergence. PSOR and BB residual definitions differ; full-budget counts alone are not stopping-tolerance tests. Two terminated APGD pilots are excluded from endpoint comparisons. No correction is qualified and no bound on exact-solution or physical error is established. These FM06-R1 results do not validate FM07-R6 or select a feeder.

### *H. Lunar and Upper-Speed Compatibility Bounds*

The lunar extension is an ideal requirements screen. An ideal coplanar, tangential patched-conic screen gives Earth departure 3.106427 km/s, lunar arrival excess speed 0.830117 km/s and direct capture into a 100 km circular lunar orbit 821.239408 m/s. Finite sphere-of-influence and epoch targeting are excluded. Capture into a $100 \times 10{,}000$ km ellipse reduces initial capture to 306.643674 m/s, but later same-perilune circularization restores the same total impulse. At the three staging examples, ideal engine-only escape thresholds are 676.620, 523.149 and 162.024 m/s. An assumed 200 m/s allowance does not establish escape from the first two. No-shot release-recoil credit is zero. Useful lifetime, navigation covariance, finite carrier burns and controlled independent disposal remain unresolved.

For a 4 kg payload and 300 kg retained host, a separate 1 km/s relative-release investigation requires at least 1.973684 MJ added relative kinetic energy and 2.012598 km constant-force stroke at the 25 g study ceiling. This rejects the compact/low-acceleration requirement combination in that ideal model; 25 g is not a payload qualification. Pusher, loss, source and catcher burdens are additional. Nine algebraic checks and 182 reproduced numerical fields support these conditional bounds. They are not provider performance or flight validation. The source package preserves the historical C0-S2 criteria-hash mismatch rather than rewriting its provenance.

### *I. Common-Boundary Four-Payload Feeder Study*

FDR09 models all seven candidates with two 1U and two 3U stored payloads and one common receiver. Fixed-cell concepts include four local feed stations and 900 mm receiver travel on 1,600 mm rails; the indexed drum retains a separate index/extraction arrangement. Passive 1U inserts

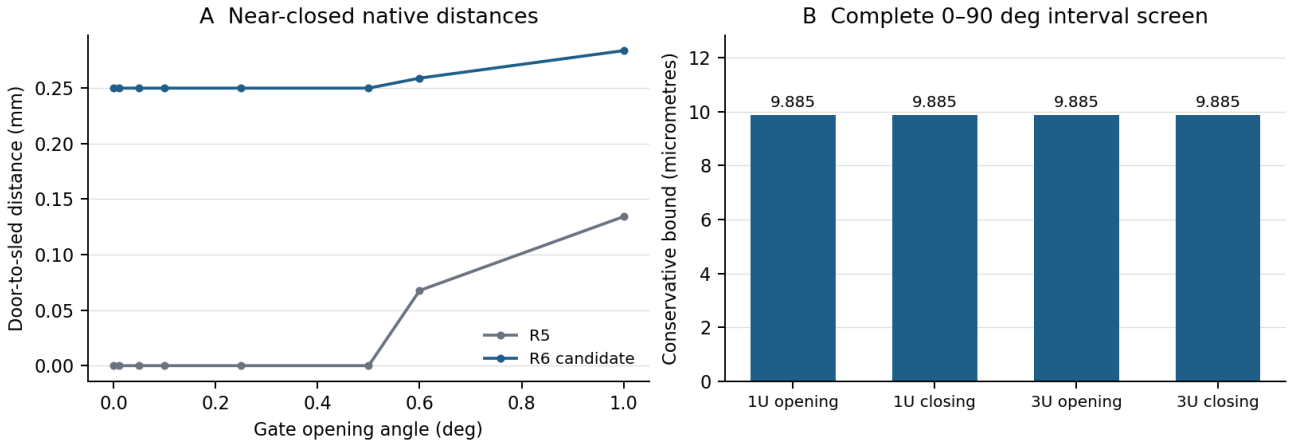



Fig. 26. RVC01 native gate/deck distances and complete-angle conservative screen for FM07-R6. The 0.25 mm panel relief and the smaller global guarded bound are different quantities. R5 near-closed zero distance denotes surface contact, not demonstrated volumetric penetration.

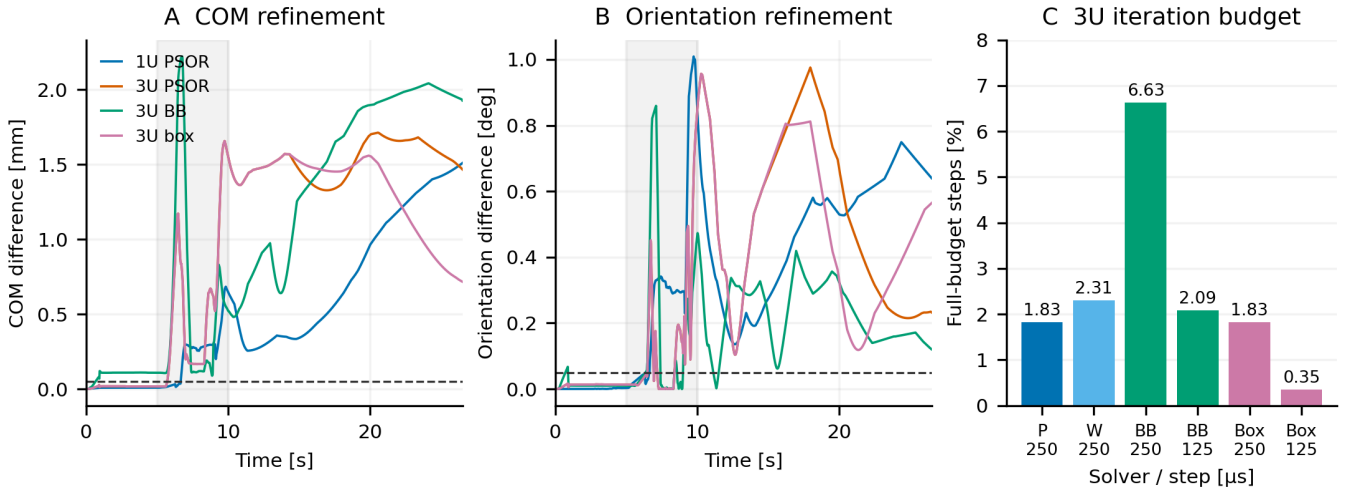



Fig. 27. FCE13 native Chrono refinement differences and 2,000-iteration budget use. P: PSOR; W: warm-start PSOR; BB: preconditioned BB; Box: equivalent box core with PSOR. Shading marks feed deceleration; dashed lines retain the 0.05 mm and 0.05° screens. A full-budget solve may meet tolerance on its last iteration. The plots measure tested numerical sensitivity rather than certified error.

TABLE IX
FDR09-R1 PARTIAL MASS ACCOUNTING [KG]. CATALOG OPTIONS ARE UNINTEGRATED; FULL INSTALLED MASS/CG/INERTIA REMAIN UNKNOWN. ALL SEVEN FAILED THE ORIGINAL FDR09 ELIGIBILITY TEST.

| Candidate | CAD material | Catalog options |
|---|---|---|
| Captive lead screw | 18.606 | 4.230 |
| Spring / dual gate | 18.915 | 6.002 |
| Rack / pawl | 18.618 | 5.830 |
| Indexed drum | 14.821 | 1.932 |
| Pantograph | 18.060 | 6.002 |
| Nested telescope | 18.132 | 4.230 |
| Belt shuttle | 17.915 | 5.830 |

preserve the common outer handling envelope and release datum; each modeled insert weighs 0.185933 kg. Positive insert lock/return custody and spacecraft-interface approval remain unestablished.

The manufacturer-data option screen contains 384 time/station cases. It does not validate a complete source-rated mechanism. A separate lead R2 trim changes the checked nominal rectangular payload sweep from eight intersections to zero and removes a 16.176779 $mm^3$ guide-bearing housing intersection. Its custom-material subtotal is 18.492547 kg. A 0.25 mm screw-bearing radial gap and unverified loaded support remain. This geometry correction does not establish full-carriage, tolerance, fault, insert-custody or installed-system acceptance. Ten native/STEP pairs have separate same-kernel-family CadQuery readback. No candidate passed all original mandatory gates in FDR09-R1; its agreed weights therefore produced no eligible downselect. The subsequent FM08-R3 study uses an explicitly revised academic-design acceptance boundary. The four-payload result does not constitute a twelve-payload integration.

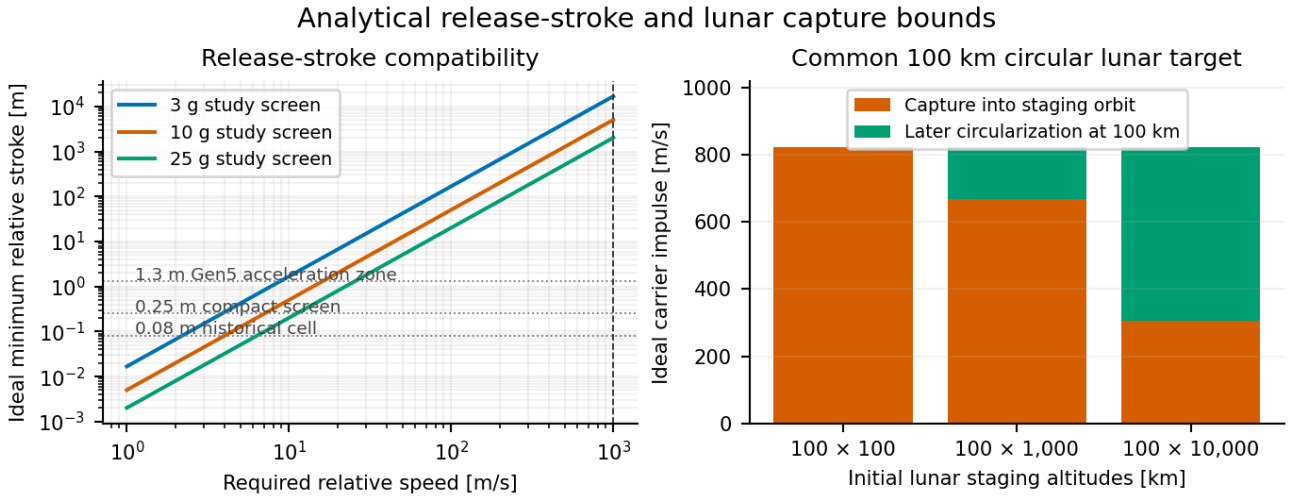



Fig. 28. Necessary release-stroke compatibility for the declared 4 kg/300 kg ideal model, and capture/circularization impulse for the same 100 km lunar target. Acceleration screens are study choices. Initial capture savings do not imply a lower total impulse or a useful long-lived staging orbit.

### *J. Selected Guided Lead-Screw Feeder: FM08-R3*

The separately controlled FM08-R3 academic design selects independent guided lead-screw cassettes and one indexed receiver for two 1U and two 3U passive payload envelopes. Fixed cassette spacers accommodate 1U length and remain in storage; retractable receiver stops distinguish the aft datums while preserving a common leading datum. The spacecraft requires no added armature or feeder attachment. This is an interface objective for the declared envelopes, not approval of a named satellite.

The stowed design contains 154 named parts and 25.364310 kg represented custom material. Adding the separate 4.473 kg primary-drive catalog subtotal gives 29.837310 kg of partial feeder accounting. Its $1{,}610\times877.8\times388$ mm bounding box is an accommodation requirement. Payload internals, omitted actuation, harness, source and launcher hardware prevent interpreting these quantities as full installed mass or host acceptance.

Five complete native assembly states each contain 154 features and 154 STEP solids, with zero inclusive positive-volume intersections. The installed matrix passes 115 sampled operating-envelope cases, with 18 additional local payload samples. Separate CadQuery/OCP imports pass all five STEP validity, solid-count and volume comparisons; the largest relative volume difference is $2.94\times10^{-14}$. Both software paths use the OpenCASCADE kernel family. These nominal geometry and serialization checks do not establish continuous tolerances, loaded contact or manufactured fit.

FDR10 applies safety/fault control, packaging/mass, positioning/custody, cycle/power and manufacturability weights of 35%, 25%, 20%, 10% and 10%. The judgment scores are 7.30 for the guided lead screw, 7.10 for rack/pawl and 6.90 for spring/two-gate transfer; belt, drum, telescope and pantograph score 6.70, 6.40, 6.05 and 5.50. The 0.20-point lead is small: equal weighting ties the top three at 7.20, packaging emphasis favors springs, and cycle emphasis favors rack/pawl. The selection therefore rests on the adopted safety-led requirements, controlled pusher displacement and continuous channel containment, rather than a unique numerical optimum or a measured reliability advantage. Rack/pawl is retained as the runner-up.

The intended sequence establishes receiver docking, empty-receiver and closed-far-stop proof, opens the storage boundary, advances the guided pusher through a capped channel, establishes seated-presence and settling proof, closes the receiver capture tabs, withdraws the pusher, closes the cell, and then permits launch preparation. A fault inhibits release and commands stop-and-contain. A retained failed cell can be isolated only after independent common-path proof and supervised recovery. A bridge obstruction inhibits the common path. The toothed-belt indexer is disconnected by a split clutch before launch arming, with independent disengagement proof. After launch, stopped-carrier proof precedes clutch engagement and engagement proof before empty return. These rules are executable logical contracts; truthful sensors, real latch engagement and brake response require physical verification.

The sizing model assumes 50 N resistance, a 100 N adverse screen and 0.50 gear efficiency. The reference 24 V motor is FAULHABER 3242G024BX4 with 32GPT gearing and MC5005S control [33], [36], [37]. Catalog continuous motor torque is 0.0418 N m. The lead-screw model includes trapezoidal-thread friction, bearing drag and reflected inertia; the receiver model includes tangential belt force at the declared 12 mm pulley radius and reflected inertia. Belt engagement, pretension and backlash are not qualified by this sizing calculation. Supplier ambient data establish conditional sizing references, without vacuum, launch or thermal qualification.

A deterministic state model passes 32,821 enumerated four-cell nominal, fault-injection, prohibited-command, recovery, isolation and repeated-release checks. A separate twelve-cell campaign passes 98,325 logic cases. Nominal payload transfer

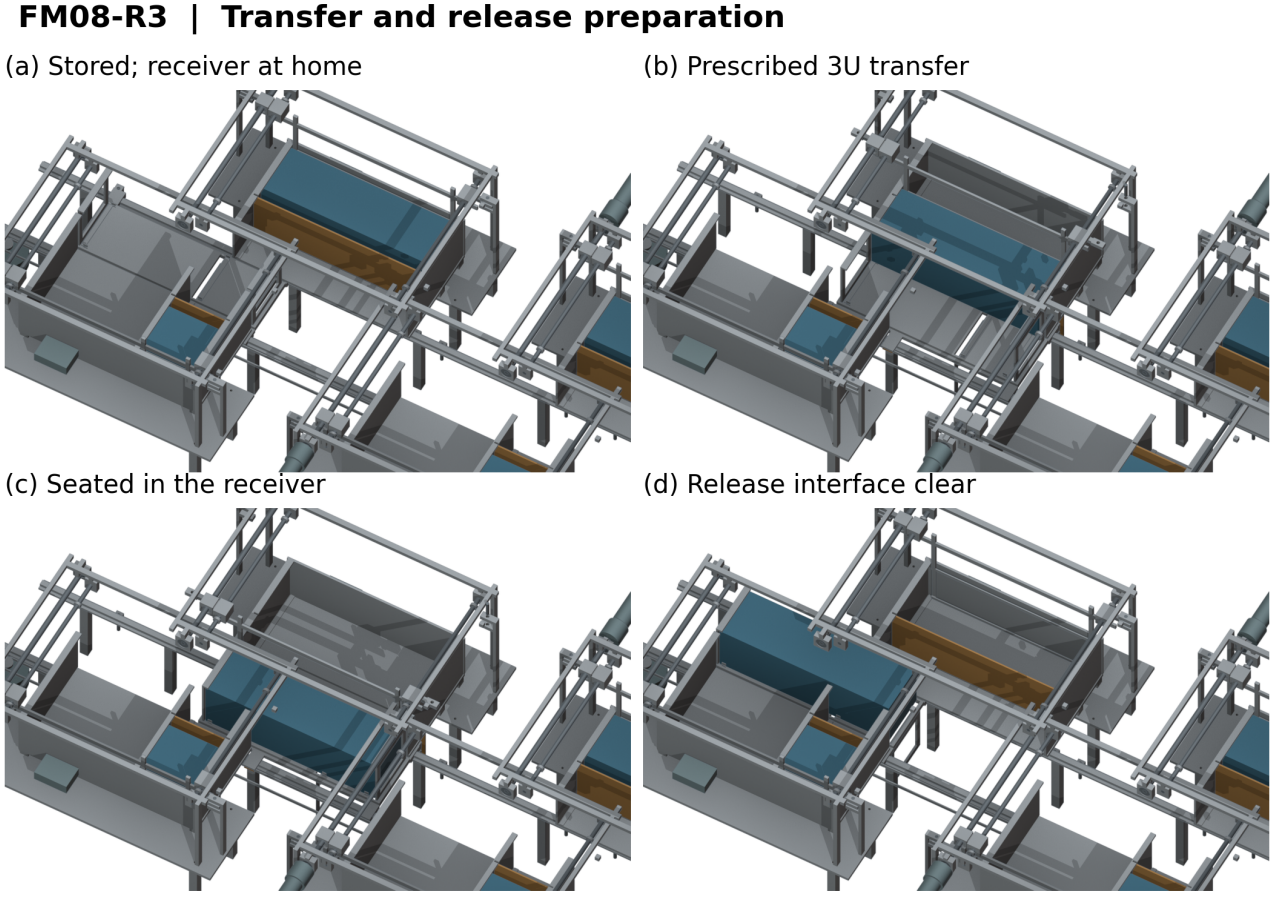



Fig. 29. FM08-R3 selected feeder in prescribed storage, transfer, seated and launch-preparation states. Visualization derived from the native CAD geometry; the sequence illustrates intended operation and is not a physical demonstration or dynamic simulation. Fixed 1U spacers remain in storage.

TABLE X
CONDITIONAL DRIVE DEMANDS FOR THE DECLARED MOTION PROFILES. THE 100 N RECEIVER CASE APPROACHES THE CATALOG TORQUE LIMIT; ASSUMED EFFICIENCY AND RESISTANCE ARE NOT MEASURED.

| Case | Torque [N m] | Speed [rpm] | Current [A] |
|---|---|---|---|
| Local / 50 N | 0.01485 | 3600 | 0.448 |
| Local / 100 N | 0.02627 | 3600 | 0.724 |
| Receiver / 50 N | 0.02038 | 4254 | 0.581 |
| Receiver / 100 N | 0.04058 | 4254 | 1.069 |

requires a coast condition with host burn off and stable-host proof; launch also requires 5 s with the burn off. An unexpected burn inhibits the cycle and resets the guard. Its containment argument uses the capped channel, closed far stop, guides, hard stops and a normally engaged static brake; screw self-locking is not assumed. The pusher does not pull the spacecraft. Low-friction forward motion and end-stop impacts are unmeasured, and reversing the pusher does not establish recovery of a jammed payload. A local analytical 0.12 N m brake reference exceeds the ideal frictionless 155 N backdrive demand by a factor of 2.43. The receiver separately uses a 2 N m static-brake reference against a 1.2 N m holding demand at 100 N and 12 mm pulley radius, a factor of 1.67. This is static sizing, not a stopping-distance or contact-impact result. The nominal corridor calculation retains 0.623 mm after the declared manufacture/alignment/thermal/75 N guide-deflection stack; it is a deterministic study screen, not a measured tolerance distribution.

FM08-R3 is selected at the computational academic-design

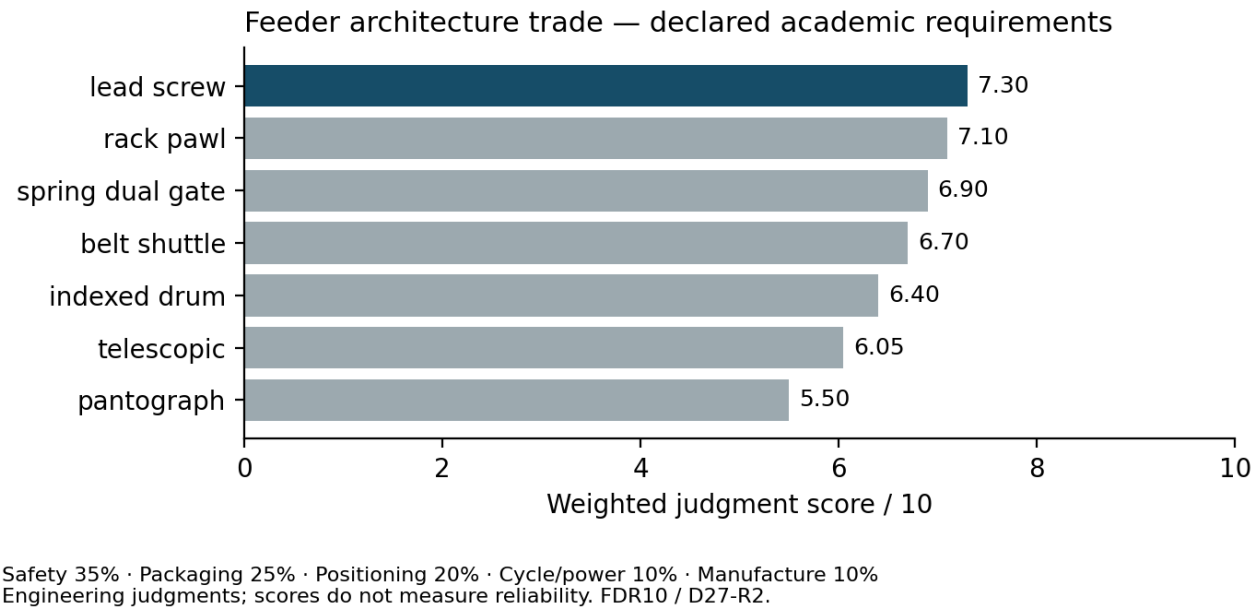



Fig. 30. FDR10 safety-weighted architecture judgments. Scores compare design features and current evidence; they are neither measured reliability nor success probabilities. Weight sensitivity changes the preferred architecture.

level. Its native geometry, STEP exchanges, calculation inputs, sequence source and raw records are supplied locally. Earlier FM06-R1 Chrono pose convergence remains unestablished and supplies no release-precision or contact-force acceptance for this revision. No Gen5 shot speed, motor rating, physical reliability, environmental qualification or provider approval transfers to the selected feeder.

### *K. Detailed Twelve-Cell Scaling Boundary*

The native twelve-cell extension contains 398 named parts on a 4,200 mm rail, with 3,500 mm receiver travel. Represented custom material is 72.730463 kg; the separate selected primary-drive catalog subtotal is 10.601 kg, yielding 83.331463 kg partial feeder accounting. The bounding envelope is 4,210×877.8×388 mm. Both exported twelve-cell states contain 398 features and 398 STEP solids; inclusive exact-solid checks find no positive-volume intersection, and all twelve modeled rectangular payload swept corridors clear represented parts. Full installed mass and provider accommodation are not established. The matched mission cases therefore add declared 10/25/50 kg omitted-equipment allowances, rather than treating unresolved hardware as massless. The twelve-cell full-index time is 77.778 s at the prescribed peak speed, with a conservative feeder-only allocation of 223.556 s per payload. These are sizing and scheduling assumptions; actual launch, braking, charging and attitude response require separate evidence.

### *L. Matched Mission Retarget with Selected Feeder Mass*

FBR-02 uses the same full-state delivery targets and common epoch for host-assisted spring separation and imposed 3–6 m/s VOLLEY release. Base host mass is 300 kg, specific impulse is 220 s and the force model includes J2. Declared limits are 100 N maneuver force and 10/30 kg initial fuel for four/twelve spacecraft, with 5 s pre-release and 20 s post-release zero-force guards. All ten targeting cases meet their stated force, fuel and residual criteria. None establishes achieved electromagnetic release speed. All 26 native Orekit replays and ten conservation/guard checks pass, with maximum nominal terminal differences of $1.973 \times 10^{-5}$ m and $2.184 \times 10^{-8}$ m/s. The 60-to-30 s step refinements change terminal position by at most 4.72 and 8.26 $\mu$m for four and twelve payloads. These are numerical consistency results. Frozen-command 3U mass probes from 4 to 6 kg miss targets by 19.543 and 43.018 km; they do not establish a robust flight accuracy envelope.

At the central 25 kg omitted-equipment allowance, four-cell VOLLEY assumes 54.837 kg device mass and uses 2.039 kg propellant at 6.979 N peak force; the 16 kg spring proxy uses 6.076 kg at 13.356 N under the same window policy. Twelve-cell VOLLEY assumes 108.331 kg and uses 4.852 kg at 51.884 N. The lowest-consumption tested twelve-cell spring policy uses 17.676 kg at 93.657 N; allowing it longer maneuver windows reduces peak force to 35.858 N while increasing consumption to 24.100 kg. Spring operation is not assigned the VOLLEY feeder handling time.

The 241.818 s twelve-cell pre-release coast exceeds the 223.556 s feeder-only allocation by 18.263 s, before complete charging, launch, braking, reset and attitude operations are resolved. At the central allowance the VOLLEY dry-device penalties are 38.837 and 60.331 kg, exceeding the corresponding 4.036 and 12.824 kg propellant-use reductions. These are conditional resource comparisons; installed-mass, optimized wet-mass, cost and customer-service advantage are

not established.

### *M. Resource and Navigation Requirement Sensitivities*

SYS-01 evaluates 108 energy/charging, 18 constant-force stroke, 36 single-axis host-impulse and 54 optimistic heat-rejection cases. A 4 kg payload at an imposed 6 m/s relative release from a 300 kg retained host requires 71.053 J of ideal relative kinetic energy. Four independent identical shots at assumed 25% net efficiency require at least 1.137 kJ bus energy and 22.74 s charging at 50 W, before additional source/catcher burdens. This homogeneous sensitivity case is distinct from the mixed-size manifest. Efficiency, charging, emissivity and deep-space sink are declared assumptions, rather than achieved hardware values.

LNS-01 adds nine two-body lunar osculating-periapsis sensitivity cases, each checked with 4,096 and 8,192 samples and a refined local state Jacobian. The middle assumed band is 100 m position, 0.1 m/s velocity, 0.1° pointing and 1% relative-speed standard deviation, taken independent and Gaussian. Linearized periapsis standard deviations are 570, 581 and 1,357 m at 100, 1,000 and 10,000 km staging apoapsides; finite-sample estimates are 573, 583 and 1,353 m. These calculations do not establish measured navigation precision or useful lunar lifetime: terrain, nonspherical gravity, external perturbations, coverage and communications are excluded. Source, numerical records and figures are included in the local system-study supplement.

### *N. Native Lunar Coast Sensitivity*

The separate LNC-01 supplement captures 18 native GMAT R2026a coast attempts [38] using DE421 Earth/Sun point perturbations and lunar gravity degree/order 20, 40 and bounded 80 comparisons. Of twelve requested 180-day cases, seven reach a 1,737.4 km spherical-radius stop and five reach the final horizon; six higher-degree/alternative-field cases reach their requested 30 days. For the 100 km circular reference at 90° to MJ2000Eq, the degree-20 and degree-40 stops occur at 115.859 and 117.984 days. At 90 days, a 120-to-60 s maximum-step change produces 0.0575 m position difference, whereas degree 20 to 40 changes it by 10.35 km. Degree 40 to 80 differences at 30 days remain 619–1,398 m across the three orbit cases. The 90° reference is not a lunar-polar plane label. These are bounded force-model sensitivities; terrain, targeting and stationkeeping are absent, and neither sphere stops nor completed coasts establish useful mission lifetime.

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